Unit 46 Factors affecting the rate of a reaction

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.



Fig. 46.1 How can she get the food cooked more quickly?

46.1 The effect of change in temperature on the rate of reaction

How can I dissolve the sugar more quickly? How can I get the food cooked more quickly? How can I get the glue to set more quickly?

The answer to all these questions probably involves raising the temperature.

Temperature has an important effect on the rate of chemical reactions and we make use of it constantly in our everyday lives. The chemical industry also depends heavily on it. As an example, there would be no Haber process for making ammonia without suitable temperature.

In Unit 45, we have seen that when dilute sulphuric acid is added to sodium thiosulphate solution, the reaction mixture becomes cloudy due to the formation of sulphur.

$$Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$$

To measure the reaction rate, we can mix certain amounts of sodium thiosulphate solution and dilute sulphuric acid in a beaker. Then record the time required for the cross to become invisible when viewed from above.

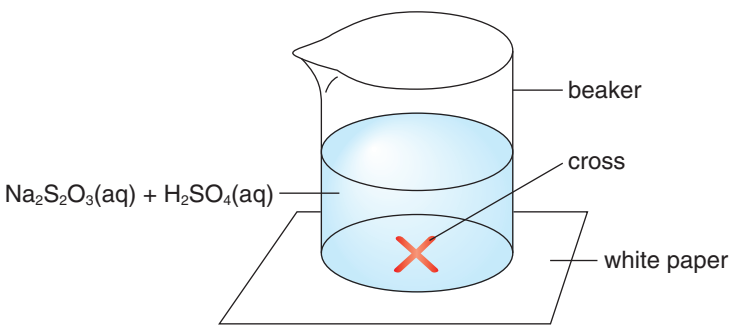


Fig. 46.2 Sodium thiosulphate solution is mixed with dilute sulphuric acid

In an experiment to investigate the effect of change in temperature on the rate of the reaction between sodium thiosulphate solution and dilute sulphuric acid, a graph shown in Fig. 46.3 is obtained.

We can see that the rate of the reaction increases rapidly with increasing temperature. The same type of results can be obtained for many reactions. How can we explain this phenomenon?

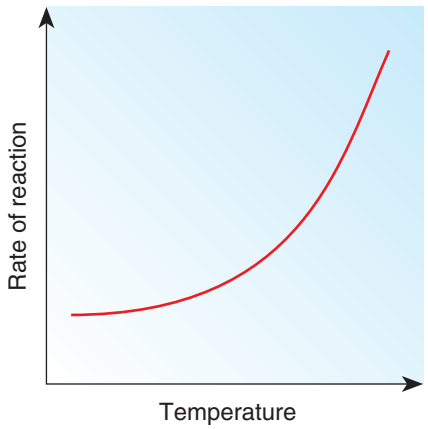


Fig. 46.3 A graph showing the effect of change in temperature on the rate of the reaction between sodium thiosulphate solution and dilute sulphuric acid

46.2 A further look at the collision theory

In Topic 10 Rate of Reaction, we have used the collision theory to explain why changes in concentration of a reactant, surface area of a solid reactant and temperature affect the rate of a reaction.

The basic ideas of the collision theory are as follows:

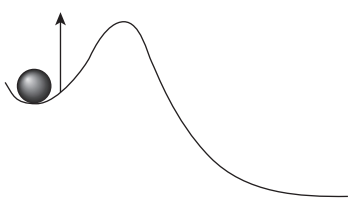
- For a collision to lead to a reaction, the particles of the reactants must collide with each other in a correct orientation.
- A reaction occurs when the particles of the reactants have a certain minimum kinetic energy.

For example, imagine two molecules, A–B and C. For a reaction to occur, the two molecules must first collide in a correct orientation.

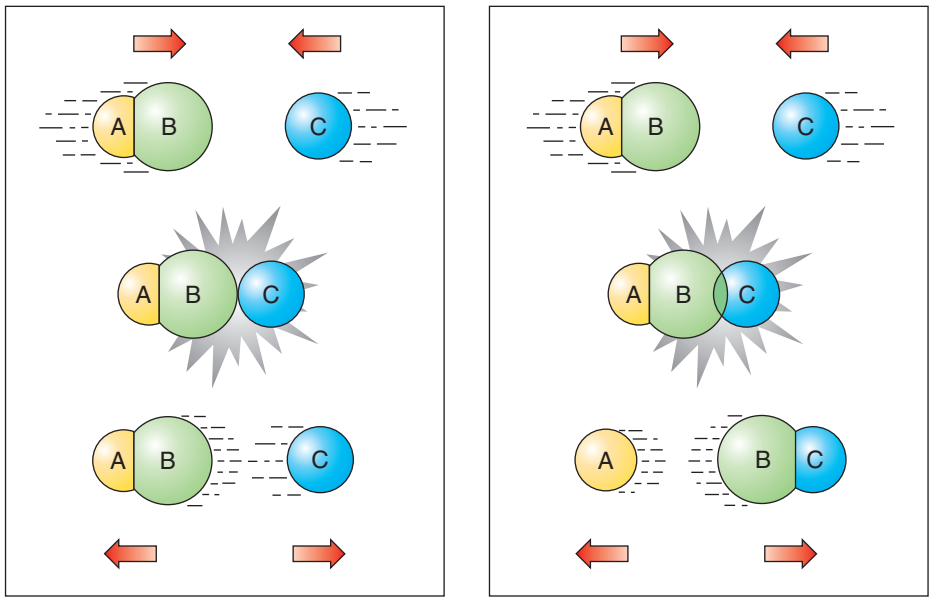
However, not every collision causes a reaction (Fig. 46.4). The existing bond starts to stretch and break and a new bond starts to form when the molecules collide. Only those pairs of molecules with combined kinetic energy equal to or greater than the **activation energy** will react to produce products*.

Activation energy is the minimum kinetic energy required by colliding particles for a reaction to occur.

* For an exothermic reaction, the situation is rather like a ball rests at the foot of a hill. A certain amount of energy has to be supplied to lift the ball over the top even though energy is given out overall when the ball rolls down the hill.



activation energy 活化能



(a) Two slow-moving molecules collide with a kinetic energy less than the activation energy. They just bounce off one another and stay unreacted.

(b) Two fast-moving molecules collide with a kinetic energy equal to or greater than the activation energy. A reaction occurs.

Fig. 46.4 For a reaction to occur, molecules colliding in a correct orientation must have kinetic energy equal to or greater than the activation energy

Energy profiles

Plots like that shown in Fig. 46.5 are a useful way of picturing how the potential energy changes as a reaction proceeds. The curve line is the energy pathway for a pair of colliding molecules. It is called the **energy profile*** for the reaction. N1

* This is an energy profile for a reaction that takes place in one single step. We will discuss energy profiles for reactions that take place in two or more steps later in this unit.

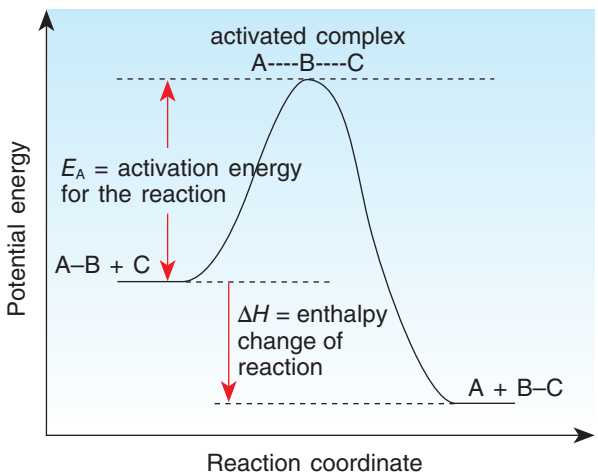


Fig. 46.5 Energy profile for an exothermic reaction A–B + C ⟶ A + B–C

Animation
Energy profile for a reaction

N1 Remind students that the x-axis of an energy profile is reaction coordinate, NOT 'reaction coordination' or 'time'.
✗ ✗

energy profile 能線圖

In going from reactants to products, the highest point on the pathway corresponds to an arrangement of atoms where old bonds are breaking and new bonds are starting to form.

For the reaction of A–B and C, it would be an arrangement in which the bond in A–B is breaking and a new bond between B and C is forming.

The highly energetic and unstable species that exists briefly at the point of maximum potential energy is called the *transition state* or *activated complex*.

Fig. 46.5 shows the energy profile for an exothermic reaction. The activation energy for the forward reaction is represented by the difference in potential energy of the activated complex and the reactants. Notice that the products have a lower potential energy than the reactants. The difference between the potential energy of the products and that of the reactants represents the enthalpy change of the reaction (ΔH).

Fig. 46.6 shows the energy profile for an endothermic reaction.

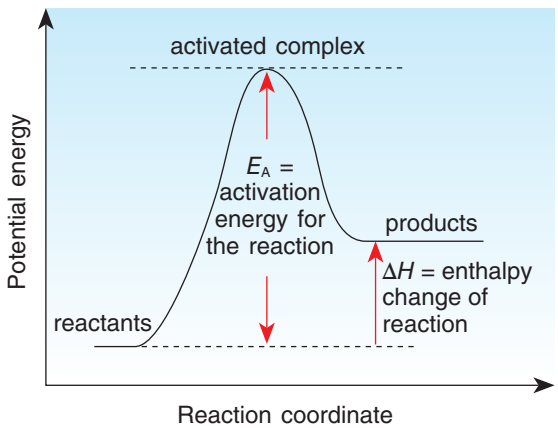


Fig. 46.6 Energy profile for an endothermic reaction

Fig. 46.7 shows the energy profile of a reversible reaction.

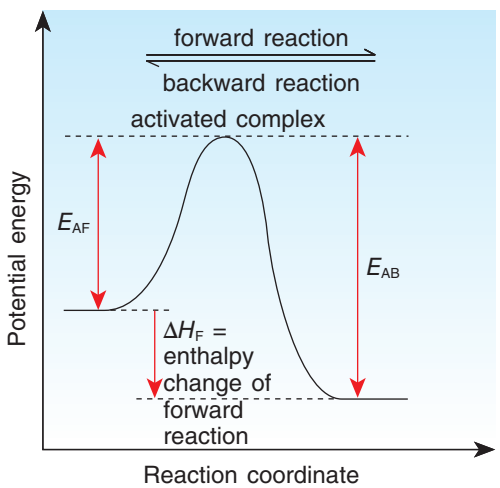


Fig. 46.7 Energy profile of a reversible reaction

transition state 過渡態　　activated complex 活化複體

A forward reaction and its backward reaction have the same activated complex.

So, if we call E_{AF} the activation energy for the forward reaction and E_{AB} the activation energy for the backward reaction, the enthalpy change of the forward reaction, ΔH_F, is

$$\Delta H_F = E_{AF} - E_{AB}$$

and the enthalpy change of the backward reaction, ΔH_B, is

$$\Delta H_B = E_{AB} - E_{AF}$$

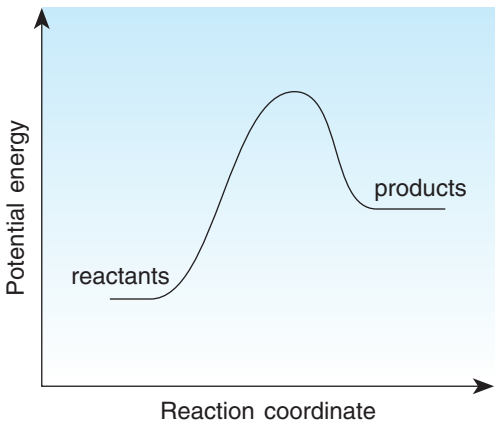
Thus, ΔH_F and ΔH_B are related to each other by

$$\Delta H_F = -\Delta H_B$$

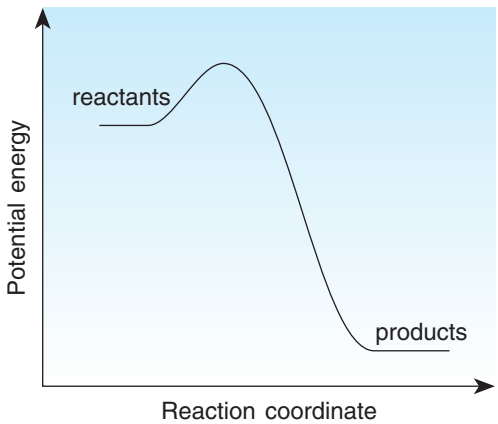
Checkpoint

The energy profiles for four different reactions are shown below. All the diagrams are drawn to the same scale.

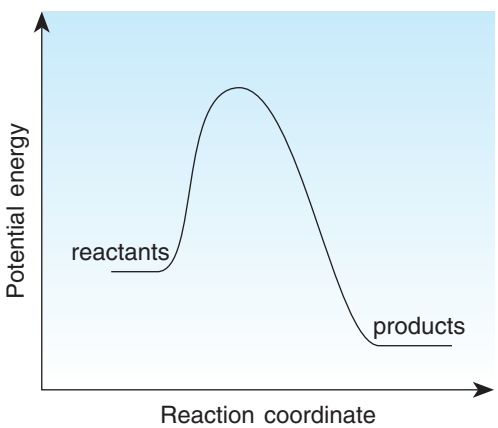
A



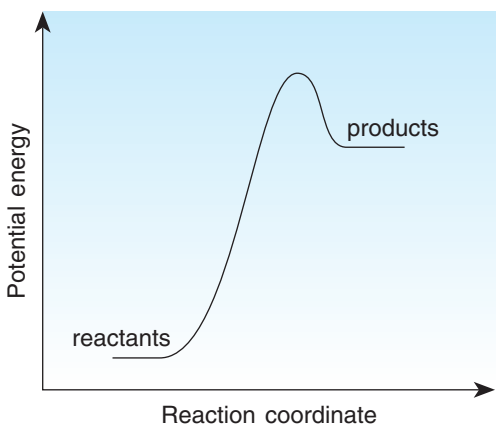
B



C



D



Which of the energy profiles

a) represent exothermic reactions?
B and C

b) represent endothermic reactions?
A and D

c) shows the largest activation energy?
D

d) shows the smallest activation energy?
B

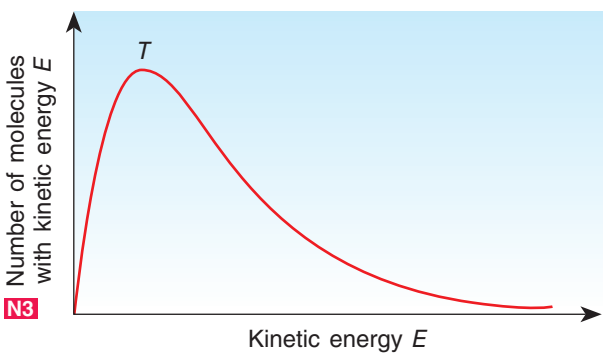
46.3 The energy distribution curve

In any gas, the molecules are constantly moving and colliding with each other. At each collision, kinetic energy is transferred between the molecules. Some collisions may lead to a gain of kinetic energy for one molecule and a loss of kinetic energy for another. A molecule may undergo a series of collisions such that it ends up with a kinetic energy higher than the average value. On the other hand, a molecule may also undergo a series of collisions such that it ends up with a kinetic energy lower than the average value. Thus, molecules of a gas at a constant temperature do not have the same kinetic energy and do not move at the same speed.

The distribution of kinetic energy among molecules in a gas is called the *Maxwell-Boltzmann distribution** of energies. The curve in Fig. 46.8 shows the number of molecules at each energy value at temperature *T*. Notice that a few molecules with very low energy values, a few with very high energy values, and most have a moderate amount of energy. The highest point on the curve shows the most probable energy. N2

* Maxwell and Boltzmann are the two scientists who predicted the results.

N2 There is no maximum energy for a molecule. The curve gets close to, but does not touch or cross, the kinetic energy axis.



N3 Notice that the y-axis of the Maxwell-Boltzmann curve may also be fraction of molecules.

Fig. 46.8 A Maxwell-Boltzmann curve for the distribution of molecular kinetic energy for a gas at temperature *T*

Maxwell-Boltzmann distribution 麥克斯韋-波爾茲曼分佈

Reference Website

Refer to the following website to explore what would happen when different concentrations and temperatures are used in a reaction:
http://phet.colorado.edu/en/simulation/reactions-and-rates
(accessed May 2015)

Effect of increasing temperature on the energy distribution curve

The area under the Maxwell-Boltzmann curve represents the total number of molecules. So the number, and therefore the area below the curve, does not change when the temperature changes.

Increasing the temperature adds energy to the system and changes the distribution of kinetic energy of the molecules.

Fig. 46.9 shows the number of molecules in a gas at each energy value at temperatures T and $T + 100$. The two curves have a similar shape but the peak of the curve '$T + 100$' moves to the right, showing that the most probable energy increases. The energies become more spread out, so there are fewer molecules with the most probable energy and the peak is slightly lower.

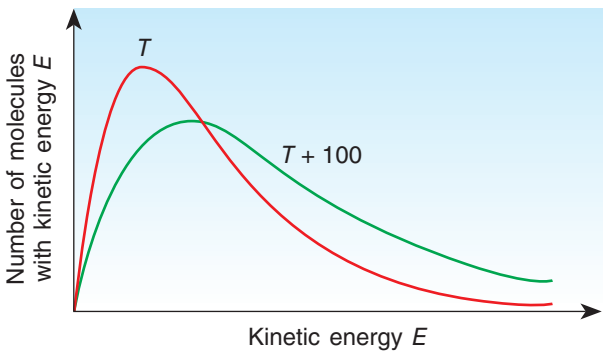


Fig. 46.9 Maxwell-Boltzmann distribution curves for a gas at temperatures T and $T + 100$

AL 2008 Paper 1 Q3(a)(iii)

Explaining the effect of temperature on the rate of reaction

Collision theory tells us that chemical reactions only occur if molecules collide with enough energy to overcome the activation energy (E_A) barrier. If we mark E_A on the Maxwell-Boltzmann distribution curve for T (Fig. 46.10), the area under the curve to the right of the line represents the number of molecules with sufficient energy to react.

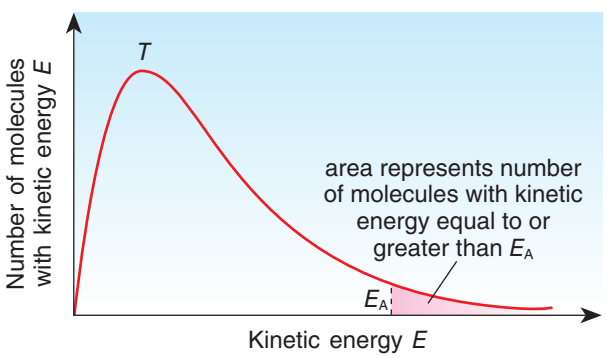


Fig. 46.10 Only molecules with kinetic energy equal to or greater than E_A can react

* Increase in temperature will also increase the collision frequency and the number of effective collisions.

Now look at the graph in Fig. 46.11, which shows the distribution curves for both T and $T + 100$. Compare the areas under the curves to the right of the line E_A. We can see that at a higher temperature, a significant larger proportion of molecules have kinetic energies equal to or greater than E_A and thus can react upon collision. This is one of the reasons for the increase in reaction rate with temperature*. N4

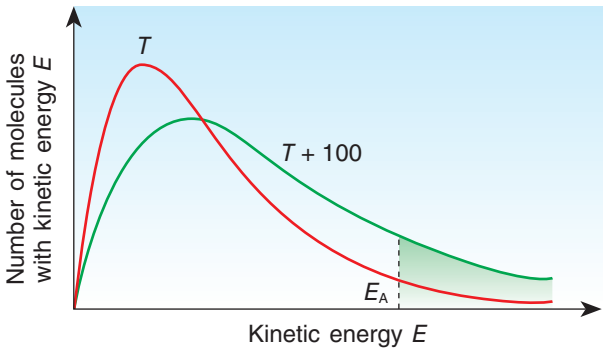


Fig. 46.11 At a higher temperature, a significant larger proportion of molecules have kinetic energies greater than E_A

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N4 Students should be able to explain why a reaction goes faster at a higher temperature in terms of the activation energy.

Increasing the temperature increases the rate of a reaction because

- with increased temperature, molecules become more energetic and move around faster and this increases the collision frequency, resulting in more effective collisions of the molecules;
- the proportion of molecules with kinetic energies equal to or greater than the activation energy becomes larger, and thus can react upon collision.

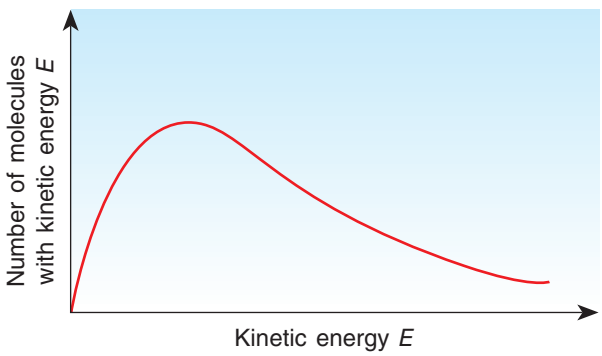
We can summarize by saying that

A reaction goes faster at a higher temperature because a larger proportion of the colliding molecules have the minimum energy needed to react.

Checkpoint

Refer to page T33.

1 The graph below shows the distribution of kinetic energy of molecules of a gas at a temperature T.



a) State ONE characteristic of the distribution of kinetic energy of molecules of the gas as shown in the above graph.

b) Sketch, on the graph, a curve to represent the distribution of kinetic energy of molecules of the gas at a temperature 'T – 100'.

c) With reference to the curves in the graph, explain how the rate of reaction changes when the temperature drops.

2 Solutions A and B are colourless. When they are mixed, they react and turn blue after a period of time. A student investigated how the change in temperature affected the rate of reaction. The rate was measured by timing how long the reaction mixture took to turn blue.

The results are shown in the table.

Temperature (°C)	22	25	34	45	51
Time for reaction mixture to turn blue (s)	290	250	200	170	160

a) i) How does the rate of reaction change as the temperature is increased?

ii) Explain why temperature has this effect on the rate of reaction.

b) State ONE variable that must be kept constant to make this experiment a fair test.

Extension

46.4 Rate equation and reaction mechanism

The energy profiles we have met so far are those for chemical reactions that take place in one single step. However, chemical reactions seldom occur in one single step. Instead, they occur in a series of steps. The series of steps for the conversion of reactants to products is known as the *reaction mechanism*.

Energy profile for a reaction which takes place in more than one step

Consider the reaction between D–E and F which takes place in two steps with the formation of an intermediate.

reaction mechanism 反應機理

Extension

Step 1	$D\text{–}E \xrightarrow{\text{slow}} \text{intermediate}$
Step 2	$\text{intermediate} + F \longrightarrow D\text{–}F + E$
Overall reaction	$D\text{–}E + F \longrightarrow D\text{–}F + E$

The slowest step in the reaction mechanism is called the *rate-determining step*.

Fig. 46.12 shows the energy profile for this reaction. The rate-determining step (i.e. D–E ⟶ intermediate) has the greater activation energy. As the energy barrier is big, few molecules have enough energy to pass over it and thus the rate of the step is low.

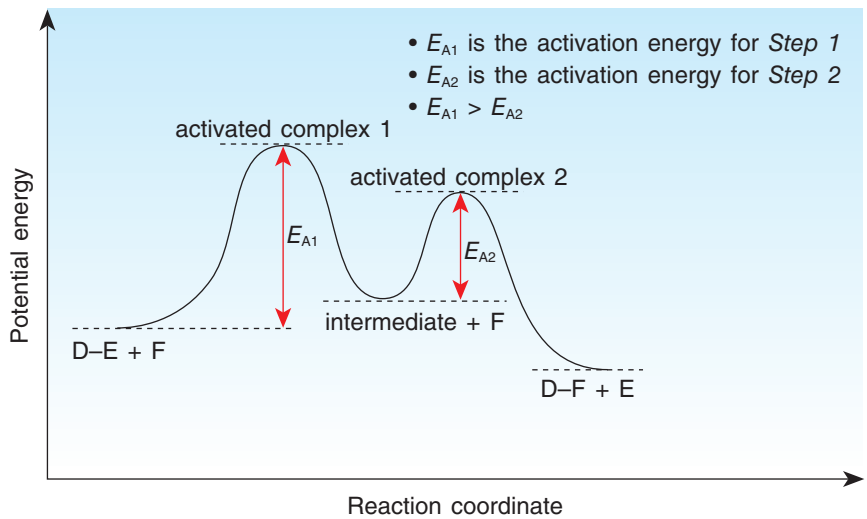
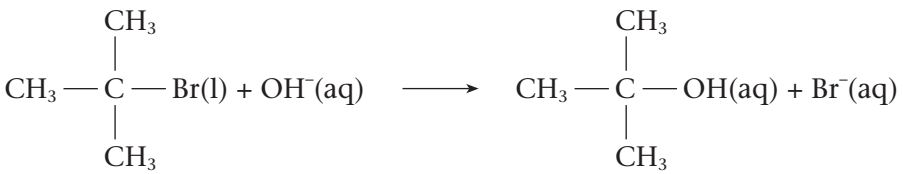


Fig. 46.12 Energy profile for a reaction which takes place in two steps

Linking the rate equation to the reaction mechanism

Once we know the rate equation for a reaction, we can link it to the reaction mechanism.

For example, consider the reaction of 2-bromo-2-methylpropane with hydroxide ion:



rate-determining step 速率控制步驟

Extension

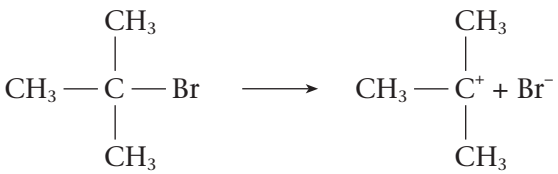
This reaction is found to be first order with respect to $(CH_3)_3CBr(l)$ and zero order with respect to $OH^-(aq)$ ion.

Rate = $k[(CH_3)_3CBr(l)]$

Why is $[OH^-(aq)]$ not involved in the rate equation? This suggests that $OH^-(aq)$ ion is not involved in the slow rate-determining step. The reaction does not take place by the direct reaction of 2-bromo-2-methylpropane itself with $OH^-(aq)$ ion. Chemists have studied this reaction in detail and found that it takes place in two steps.

First, the C–Br bond breaks in the following way:

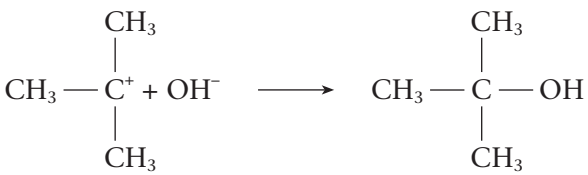
Step 1



Because this step only involves 2-bromo-2-methylpropane, its rate depends on $[(CH_3)_3CBr(l)]$, but not $[OH^-(aq)]$.

The second step involves the reaction of the carbocation, $(CH_3)_3C^+$, with the OH^- ion.

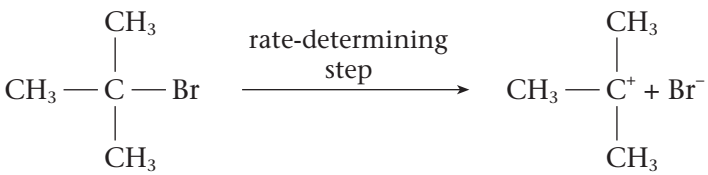
Step 2



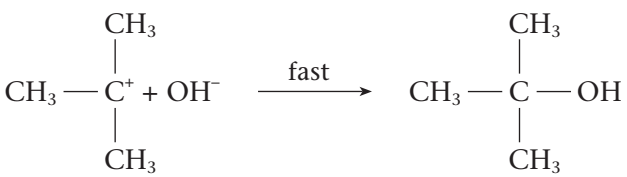
Like most ionic reactions, this process is very fast, certainly faster than *Step 1*.

The reaction can be summarized as follows:

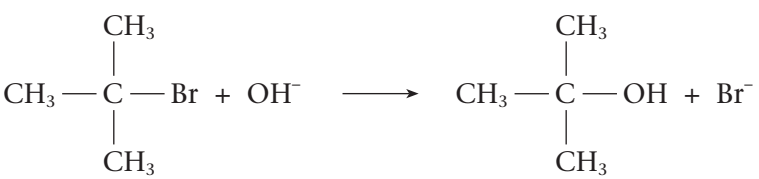
Step 1 rate-determining step



Step 2 fast



Overall reaction



Extension

So, the rate of *Step 1* controls the rate of the overall reaction. That is why the rate of reaction depends only on the concentration of 2-bromo-2-methylpropane and [OH^-(aq)] does not appear in the rate equation.

Fig. 46.13 shows the energy profile for the reaction of 2-bromo-2-methylpropane with hydroxide ion. *Step 1* (the rate-determining step) has the greater activation energy.

In any reaction with several steps, the rate-determining step is the one with the greatest activation energy.

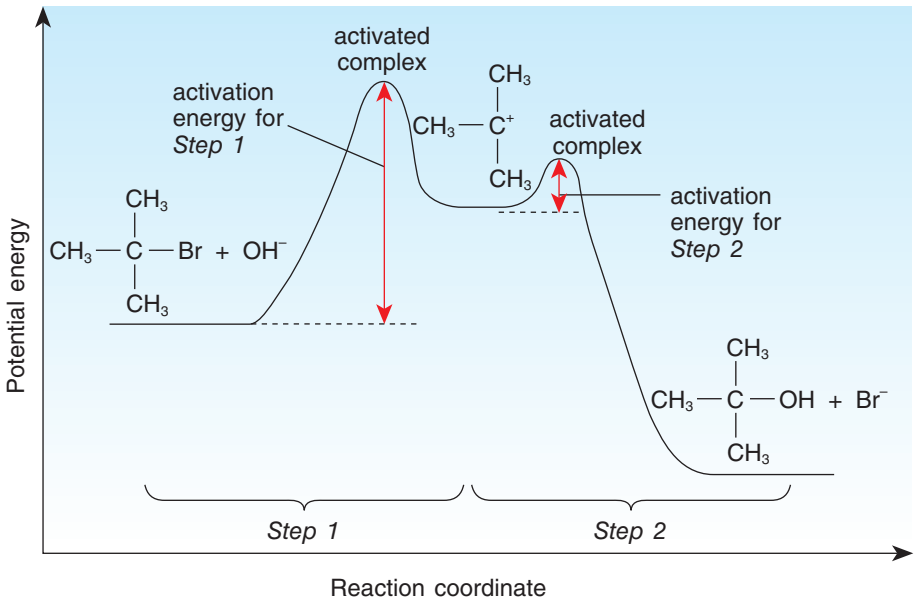


Fig. 46.13 The energy profile for the reaction of 2-bromo-2-methylpropane with hydroxide ion

Checkpoint

Refer to page T33.

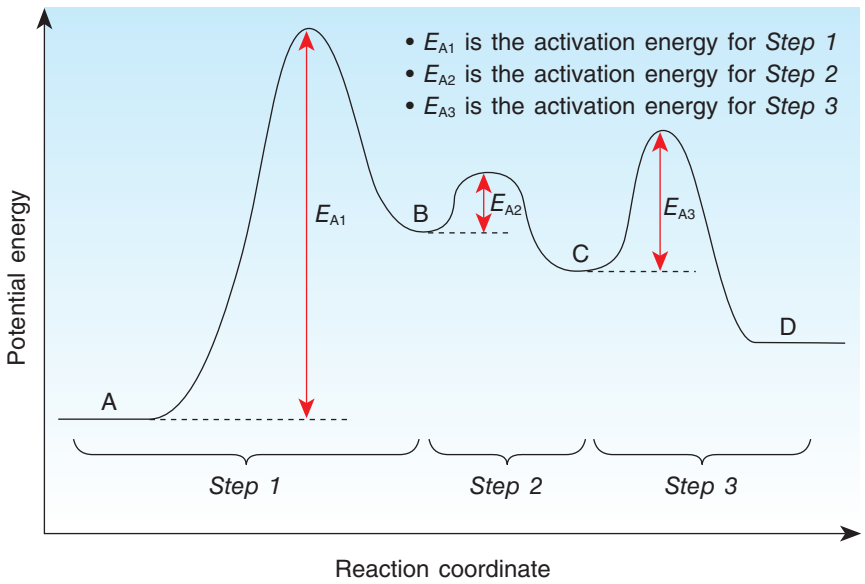
1 The reaction of bromoethane and hydroxide ion takes place in one single step.

$$CH_3CH_2Br(l) + OH^-(aq) \longrightarrow CH_3CH_2OH(aq) + Br^-(aq) \qquad \Delta H < 0$$

Draw an energy profile for this reaction. Show the energy level of the activated complex on the profile.

Extension

2 Consider the following energy profile for the reaction of A ⟶ D.



a) Explain the meaning of activation energy.

b) How many intermediates are formed in the reaction?
2

c) Which step is rate-determining? Explain your answer.

d) Is the overall reaction exothermic or endothermic? Explain your answer.

3 Consider the reaction between propanone and iodine:

$$CH_3COCH_3(aq) + I_2(aq) \xrightarrow{H^+(aq)} CH_3COCH_2I(aq) + HI(aq)$$

The rate equation for the reaction is as follows:

$$\text{Rate} = k[CH_3COCH_3(aq)][H^+(aq)]$$

a) State the overall order of the reaction.
2

b) Suggest, with reasons, which of the following two possible mechanisms fits the rate equation.

Mechanism 1
$CH_3COCH_3 + H^+ \longrightarrow$ intermediate (slow)
intermediate $+ I_2 \longrightarrow$ products (fast)

Mechanism 2
$CH_3COCH_3 + I_2 \longrightarrow$ intermediate (slow)
intermediate $+ H^+ \longrightarrow$ products (fast)

46.5 The Arrhenius equation

Consider the Maxwell-Boltzmann curve for the distribution of molecular kinetic energy for a gas at temperature T again (Fig. 46.14).

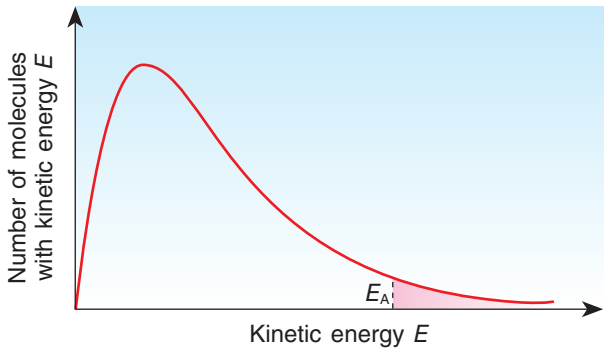


Fig. 46.14 Maxwell-Boltzmann curve for the distribution of molecular kinetic energy for a gas at temperature T

The area beneath the curve is proportional to the total number of molecules involved. Furthermore, the number of molecules with energy equal to or greater than E_A is proportional to the red area beneath the curve.

Hence the fraction of molecules with energy equal to or greater than E_A is given by the ratio

$$\left(\frac{\text{red area beneath curve}}{\text{total area beneath curve}}\right)$$

N5 Refer to page T34.

According to Maxwell and Boltzmann, the fraction of molecules N5 with an energy equal to or greater than E_A in J mol^{-1} is given by

$$e^{-\frac{E_A}{RT}}$$

where R is the ideal gas constant and T is the temperature in Kelvin.

This suggest that at a given temperature T, the rate of reaction is proportional to $e^{-\frac{E_A}{RT}}$. Now, since the rate constant, k, is a measure of the rate, we can write

$$k \propto e^{-\frac{E_A}{RT}}$$

i.e. $k = Ae^{-\frac{E_A}{RT}}$

Reference Website

Refer to the following website on a virtual experiment showing how the rate of consumption of a species varies with the activation energy:
http://www.chm.davidson.edu/ronutt/che115/Kinetics/Kinetics.htm
(accessed May 2015)

where k is the rate constant of the reaction;
A is a constant which is independent of temperature;
e is the base of the natural logarithm;
E_A is the activation energy for the reaction in J mol^{-1};
R is the ideal gas constant (i.e. 8.31 J K^{-1} mol^{-1}); and
T is the temperature in kelvin.

The last expression is called the *Arrhenius equation* because it was first predicted by the Swedish chemist Svante Arrhenius.

From the equation, we can see that raising temperature T makes $\frac{E_A}{RT}$ smaller, then k will increase and the reaction will proceed faster. Conversely, lowering temperature T makes $\frac{E_A}{RT}$ larger, then k will decrease and the reaction will proceed more slowly. Fig. 46.15 shows the relationship between rate constant and temperature.

Similarly, if E_A is low, $\frac{E_A}{RT}$ will be small. Then, k will be large and the reaction will be fast. If E_A is high, $\frac{E_A}{RT}$ will be large. Then, k will be small and the reaction will be slow.

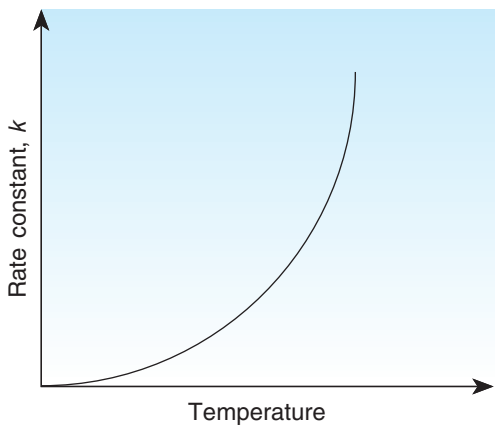


Fig. 46.15 A graph showing the relationship between rate constant, k, and temperature

N6 Refer to page T35.

The mathematics is easier if we take the logarithm of both sides of the Arrhenius equation:

$$\ln k = \ln A - \frac{E_A}{RT}$$

$$\text{or } \log k = \log A - \frac{E_A}{2.3R} \times \frac{1}{T} \text{ *}$$

So, a graph of log k against $\frac{1}{T}$ will give a straight line of slope $-\frac{E_A}{2.3R}$ *. We can calculate the activation energy, E_A, from the slope (see Examples 46.1 and 46.2). N6

* Notice that

$$\log x = \frac{\ln x}{\ln 10} = \frac{\ln x}{2.3}$$

* The equation

$$\log k = -\frac{E_A}{2.3R} \times \frac{1}{T} + \log A$$

is in the form of $y = mx + c$.

Arrhenius equation 阿列紐斯方程式

Example

46.1 Determining the activation energy and rate constant for a reaction

For the reaction:

$$2HI(g) \longrightarrow I_2(g) + H_2(g)$$

the values of rate constant k at different temperatures are obtained:

T (K)	k ($dm^3\ mol^{-1}\ s^{-1}$)
633	1.78×10^{-5}
666	1.07×10^{-4}
697	5.01×10^{-4}
715	1.05×10^{-3}
781	1.51×10^{-2}

AL 2000 Paper 2 Q3(b)(ii)

a) Plot a graph of log k on the vertical axis against $\frac{1}{T}$ on the horizontal axis.

b) Calculate the activation energy, E_A, for the reaction. DSE 2013 Paper 2 Q1(a)(ii)

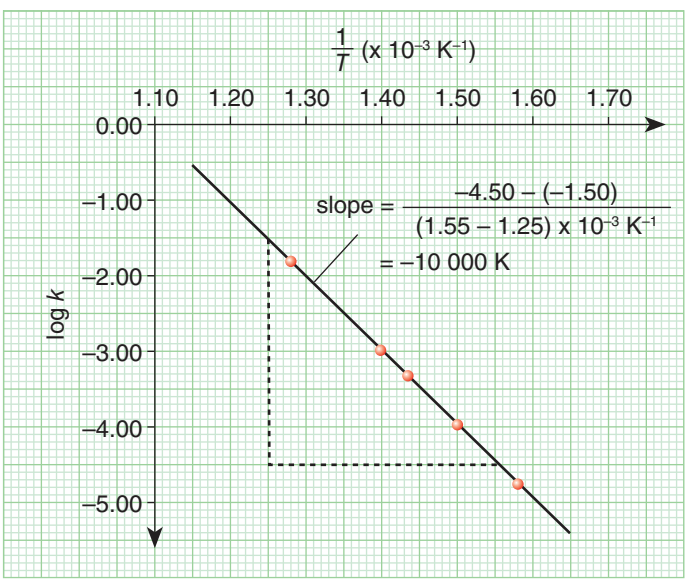
c) Estimate the rate constant of the reaction at 851 K.

(Ideal gas constant R = 8.31 J K^{-1} mol^{-1})

Solution

a)

log k	$\frac{1}{T}$ ($\times 10^{-3}\ K^{-1}$)
–4.75	1.58
–3.97	1.50
–3.30	1.43
–2.98	1.40
–1.82	1.28



b) Slope $= -\dfrac{E_A}{2.3R} = \dfrac{-4.50 - (-1.50)}{(1.55 - 1.25) \times 10^{-3}\ K^{-1}} = -10\ 000\ K$

$E_A = 2.3 \times 8.31\ J\ K^{-1}\ mol^{-1} \times 10\ 000\ K$

$= 191\ 000\ J\ mol^{-1}$

$= 191\ kJ\ mol^{-1}$

c) From the graph, at 851 K, log k = –0.800, i.e. $k = 0.158\ dm^3\ mol^{-1}\ s^{-1}$.

∴ the activation energy for the reaction is 191 kJ mol^{-1} and the rate constant at 851 K is 0.158 $dm^3\ mol^{-1}\ s^{-1}$.

46.2 Determining the activation energy for a reaction

Consider the reaction of bromide ion and bromate ion in acidic solution:

$$5Br^-(aq) + BrO_3^-(aq) + 6H^+(aq) \longrightarrow 3Br_2(aq) + 3H_2O(l)$$

AL 2000 Paper 1 Q7(b)

In an experiment to determine the activation energy for the reaction, a student prepared several reaction mixtures of the same composition at different temperatures. The time t taken for the formation of a small, fixed amount of bromine from each reaction mixture was measured. Results of the experiment are shown below: N7

T (°C)	30	40	50	60	70
t (s)	50	27	15	8	?

a) The student used a colorimeter* to follow the progress of the reaction. Describe and explain how the time t can be found experimentally.

b) Determine the activation energy for the reaction by plotting an appropriate graph.

c) Estimate the value of t at 70 °C.

(Ideal gas constant $R = 8.31\ J\ K^{-1}\ mol^{-1}$)

* We have discussed using a colorimeter to follow the progress of a reaction in Topic 10 Rate of Reaction.

Solution

a) The colour intensity of the reaction mixture increases as the yellow-brown bromine is formed. The elapsed time for each reaction mixture to reach a certain level of absorbance (or colour intensity) indicates the time t taken for the formation of a certain amount of bromine in the reaction mixture.

N7 Measurements to be taken in order to determine the activation energy of a reaction
The concentrations of the reactants should remain unchanged. The experiment is carried out at different temperatures to determine the corresponding initial rates.

b) Now reaction rate $\propto \frac{1}{t}$

$\therefore k = \frac{c}{t}$

where c is a constant. Hence

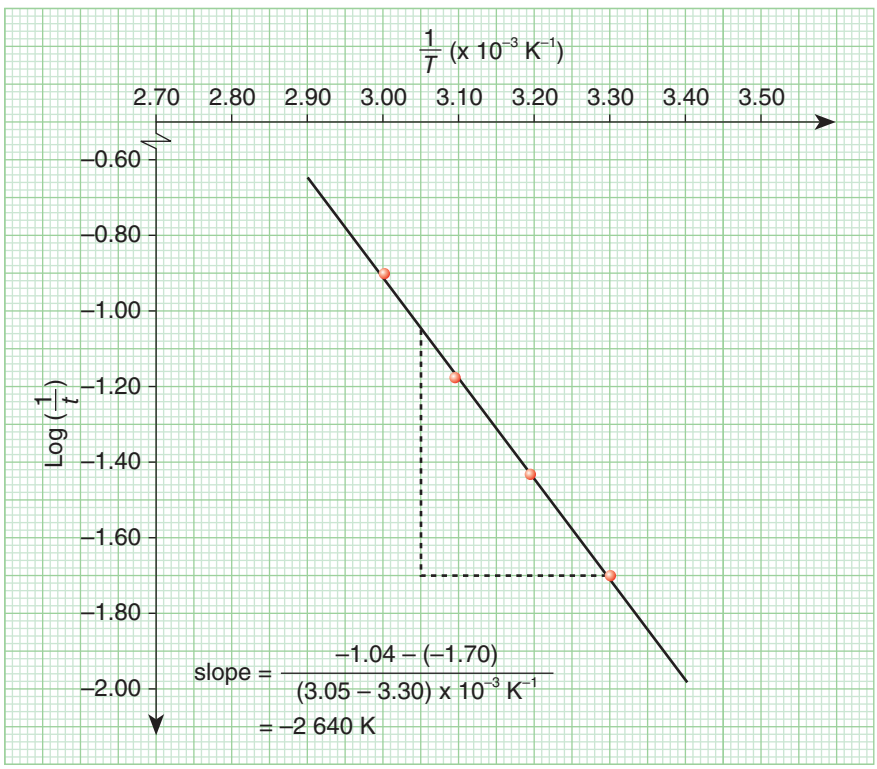
$\log (\frac{c}{t}) = \log A - \frac{E_A}{2.3R} \times \frac{1}{T}$

$\log c + \log \frac{1}{t} = \log A - \frac{E_A}{2.3R} \times \frac{1}{T}$

$\log \frac{1}{t} = \log A - \frac{E_A}{2.3R} \times \frac{1}{T} - \log c$

So, a graph of $\log \frac{1}{t}$ against $\frac{1}{T}$ has a slope of $-\frac{E_A}{2.3R}$.

t (s)	$\log (\frac{1}{t})$	T (K)	$\frac{1}{T}$ (x 10^{-3} K^{-1})
50	–1.70	303	3.30
27	–1.43	313	3.19
15	–1.18	323	3.10
8	–0.903	333	3.00



Slope $= -\frac{E_A}{2.3R} = \frac{-1.04 - (-1.70)}{(3.05 - 3.30) \times 10^{-3}\ K^{-1}} = -2\ 640\ K$

$E_A = 2.3 \times 8.31\ J\ K^{-1}\ mol^{-1} \times 2\ 640\ K = 50\ 500\ J\ mol^{-1} = 50.5\ kJ\ mol^{-1}$

c) From the graph, at 343 K,

$\log (\frac{1}{t}) = -0.68$

$\therefore t = 4.8\ s$

In the following activity, you are going to determine the activation energy for a reaction.

Video
Determining the activation energy for a reaction

Assessment Form
An assessment form on this activity is available for download from *Jing Kung Chemistry Website*.

Activity 46.1

Determining the activation energy for a reaction.

We can use an equation relating the rate constants k_1 and k_2 at temperatures T_1 and T_2 to calculate the activation energy or to find the rate constant at another temperature if the activation energy is known. To derive such an equation, we start with

$$\log k_1 = \text{constant} - \frac{E_A}{2.3R} \times \frac{1}{T_1}$$

$$\log k_2 = \text{constant} - \frac{E_A}{2.3R} \times \frac{1}{T_2}$$

Subtracting $\log k_1$ from $\log k_2$ gives

$$\log k_2 - \log k_1 = \frac{E_A}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log \frac{k_2}{k_1} = \frac{E_A}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

N8 Examination questions may give the values of k, A and T, and ask students to calculate E_A.

Example

46.3 Calculating the activation energy for a reaction

The rate constant of a certain reaction at 337 K is found to be 8 times of that at 318 K. Calculate the activation energy for the reaction. (Ideal gas constant R = 8.31 J K^{-1} mol^{-1}) **N8**

DSE 2015 Paper 2 Q1(a)(iii)
AL 2004 Paper 2 Q3(a)(iii)

Solution

$$k = Ae^{-\frac{E_A}{RT}}$$

$$\log \frac{8k}{k} = \frac{E_A}{2.3R}\left(\frac{1}{318} - \frac{1}{337}\right)$$

$$\log 8 = \frac{E_A}{2.3R}\left(\frac{1}{318} - \frac{1}{337}\right)$$

$$E_A = 97.4 \text{ kJ mol}^{-1}$$

$\therefore$ the activation energy for the reaction is 97.4 kJ mol^{-1}.

46.4 Predicting the effect of temperature increase on rate constants

Consider the following system, which comprises two single step reactions.

$$2X(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} Y(g) + Z(g) \qquad \Delta H > 0$$

(k_1 and k_{-1} are the rate constants.)

a) Write the respective rate equations for the forward and backward reactions.

b) Sketch a labelled energy profile for the forward reaction.

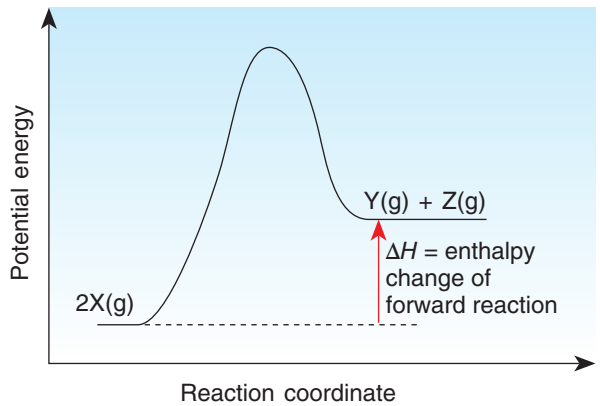
c) Predict, with explanation, whether k_1 or k_{-1} will increase to a greater extent when the temperature of the system is increased.

Solution

a) Forward rate = $k_1[X(g)]^2$

Backward rate = $k_{-1}[Y(g)][Z(g)]$

b)



c) The following equation relates the rate constants k_{T1} and k_{T2} of a reaction at two temperatures T_1 and T_2.

$$\log \frac{k_{T2}}{k_{T1}} = \frac{E_A}{2.3R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

Refer to the system in the equation, the activation energy (E_A) for the forward reaction is greater than that of the backward reaction. Therefore when the temperature is increased, the rate constant of the forward reaction (k_1) will increase to a greater extent.

Checkpoint

Refer to page T36.

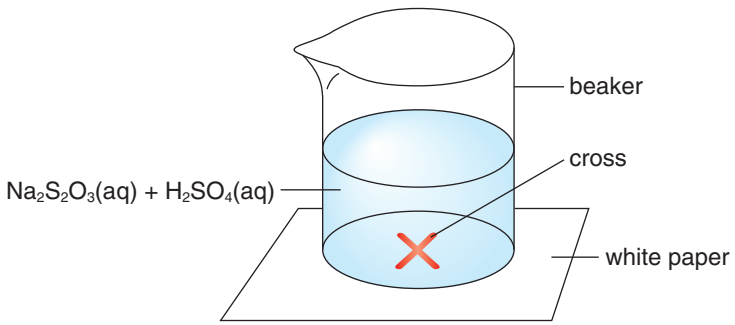
1 When bromoethane and concentrated ammonia solution are heated in a sealed tube for 15 minutes, the following reaction takes place:

$$C_2H_5Br(l) + 2NH_3(aq) \longrightarrow C_2H_5NH_2(aq) + NH_4Br(aq)$$

This reaction is very slow at room temperature. Comment on the value of the activation energy compared with that of a reaction which takes place rapidly at room temperature.

2 An experiment was carried out to determine the activation energy for the reaction:

$$Na_2S_2O_3(aq) + H_2SO_4(aq) \longrightarrow Na_2SO_4(aq) + SO_2(g) + H_2O(l) + S(s)$$



The time *t* required for the cross to become invisible when viewed from above was measured at different temperatures. The results are shown in the table below:

Temperature (K)	287	297	307	317	327	337
***t* (s)**	229	126	69	37	23	?

a) Determine the activation energy for the reaction by plotting an appropriate graph.
45.9 kJ mol^{-1}

b) Estimate the value of *t* at 337 K.
14

(Ideal gas constant R = 8.31 J K^{-1} mol^{-1})

3 The rate constant of a certain reaction at 25 °C was found to be 4.25 x 10^{-4} s^{-1} while that at 100 °C was 59.0 s^{-1}. What is the activation energy for the reaction?
146 kJ mol^{-1}

(Ideal gas constant R = 8.31 J K^{-1} mol^{-1})

46.6 Catalysts

ASL 2006 Q10(a)(ii)

A catalyst alters the rate of a reaction without itself undergoing any permanent chemical changes. Catalysis is the action of the catalyst on the reaction. N9

There are two types of catalysts: *positive catalysts* and *negative catalysts*. A positive catalyst is one that speeds up a reaction while a negative catalyst is one that slows down a reaction.

Catalysts play a crucial role in the lives of everybody. Biological catalysts called enzymes dramatically increase the rates of thousands of reactions taking place in our bodies. Catalysts are also involved in the manufacture of most chemicals that we rely on, for example, the manufacture of ammonia from nitrogen and hydrogen in the Haber process.

Catalysts are not used up in chemical reactions and we can always get them back at the end. They do not undergo any permanent chemical changes, though sometimes they may be changed physically. For example, the surface of a solid catalyst may crumble or become roughened. This suggests that the catalyst is taking some parts in the reaction, but is being regenerated.

Only small amounts of catalyst are usually needed. A catalyst does not affect the amount of product formed, but only the rate at which a product is formed.

Another interesting feature of catalysts is that they are often specific for one particular reaction. This is particularly true for enzymes.

A catalyst speeds up both the forward and backward reactions of a reversible reaction. However, it has no effect on the position of equilibrium and does not affect the yield. What the catalyst does is to decrease the time required for the system to reach equilibrium.

N9 Examples of catalysts used in different reactions

The table below lists examples of catalysts used in different reactions.

Catalyst	Reaction
Iron	Haber process in the manufacture of ammonia $N_2(g) + 3H_2(g) \overset{Fe}{\rightleftharpoons} 2NH_3(g)$
Platinum or vanadium(V) oxide	Contact process in the manufacture of sulphuric acid $2SO_2(g) + O_2(g) \overset{Pt \text{ or } V_2O_5}{\rightleftharpoons} 2SO_3(g)$
Rhodium	Conversion of carbon monoxide and nitrogen monoxide to carbon dioxide and nitrogen in catalytic converters in vehicles $2CO(g) + 2NO(g) \xrightarrow{Rh} 2CO_2(g) + N_2(g)$
Manganese(IV) oxide	Decomposition of hydrogen peroxide $2H_2O_2(aq) \xrightarrow{MnO_2} 2H_2O(l) + O_2(g)$
Concentrated sulphuric acid	Esterification reaction between ethanoic acid and ethanol $CH_3COOH(l) + CH_3CH_2OH(l) \overset{\text{conc. } H_2SO_4}{\rightleftharpoons} CH_3COOCH_2CH_3(l) + H_2O(l)$

positive catalyst 正催化劑　　negative catalyst 負催化劑

DSE 2012 Paper 2 Q1(a)(i)(2)
DSE 2013 Paper 2 Q1(b)(i)
ASL 2000 Q14
ASL 2005 Q13
ASL 2006 Q10(b)(ii)
AL 2000 Paper 1 Q9

N10 Important parameters that impact on the greenness of a catalyst

There are three important parameters that impact on both the commercial viability and the greenness of a catalyst:

1 Selectivity — the percentage of consumed reactant that is converted to the desired product (a catalyst will be of limited benefit if it also enhances the rate of by-product formation).
2 Turnover frequency — the number of moles of product produced per mole of catalyst per second (low turnover frequencies will mean large amounts of catalyst are required, resulting in higher cost and potentially more waste).
3 Turnover number — the amount of product per mole of catalyst (this is related to catalyst lifetime and hence to cost and waste).

Reference:

Mike Lancaster (2002). Green Chemistry. Royal Society of Chemistry.

46.7 How does a catalyst work N10

A catalyst works by providing an alternative pathway for the reaction to take place. In case of a positive catalyst, it provides an alternative pathway with a lower activation energy. As the activation energy is smaller, there is a greater fraction of molecules with the minimum kinetic energy needed to react, and hence the reaction goes faster.

On the other hand, a negative catalyst works by providing an alternative pathway with a higher activation energy. As a result, there is a smaller fraction of molecules with the minimum kinetic energy needed to react, and hence the reaction is slower.

ASL 2006 Q10(b)(i)

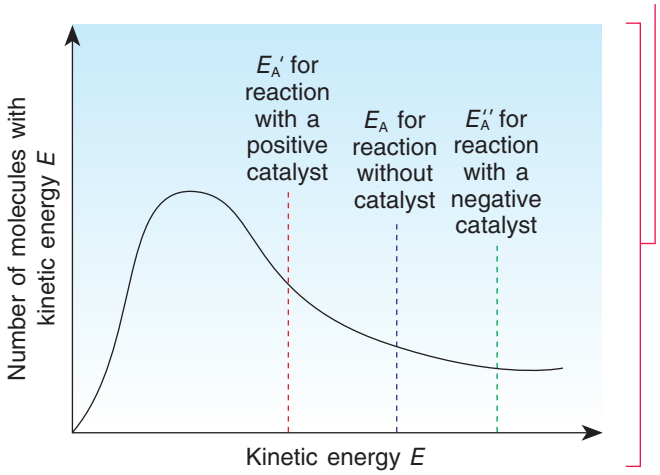


Fig. 46.16 Effect of catalysts on the fraction of molecules with the minimum kinetic energy needed to react N11

N11 Refer to page T38.

N12 Refer to page T38.

In general, a catalyst is chemically involved in a reaction. It is consumed in one reaction step and regenerated in a subsequent step. N12

Step 1	A + catalyst + B ⟶ A–catalyst (intermediate)
Step 2	A–catalyst + B ⟶ A–B + catalyst
Overall reaction	A + B ⟶ A–B

Fig. 46.17 shows the energy profiles for an uncatalyzed reaction and two catalyzed reactions (for a positive catalyst and a negative catalyst). Notice that the enthalpy change (ΔH) is the same for the uncatalyzed and catalyzed reactions.

ASL 2000 Q14
ASL 2005 Q13
AL 2000 Paper 1 Q9

DSE 2015 Paper 2 Q1(b)(i)(1)
ASL 2003 Q4(a)(ii)
ASL 2011 Paper 2 Q4(c)(ii)
AL 2001 Paper 2 Q3(a)(iii)
AL 2008 Paper 2 Q2(b)(iii)
AL 2011 Paper 2 Q1(c)(ii)

N13 Students should be able to draw the energy profiles of uncatalyzed and catalyzed reactions. Make sure that E_A (catalyzed) is less than E_A (uncatalyzed).

N14 Catalysts normally work by forming an intermediate with the reactants. The intermediate then breaks down to give the products and reform the catalyst. That is why the energy profile for a catalyzed reaction has two humps — one for each step.

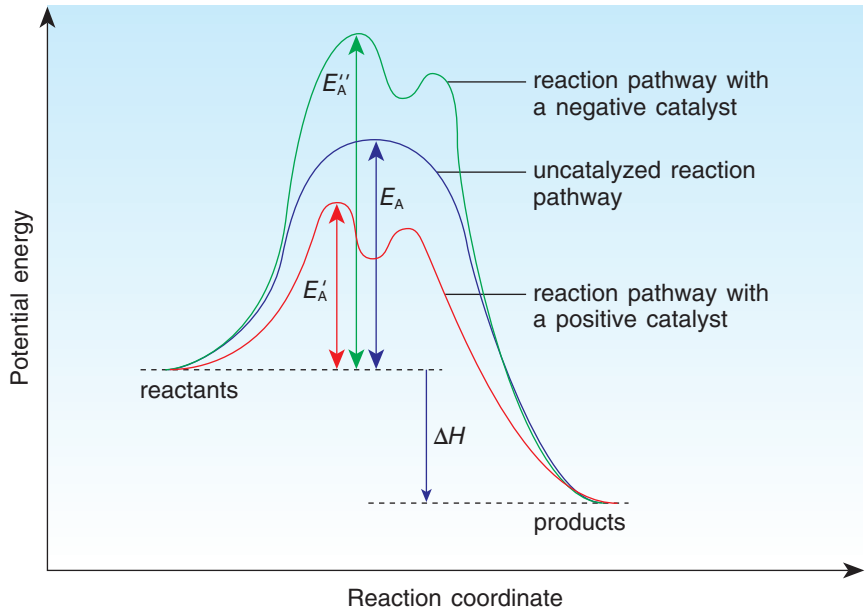


Fig. 46.17 Energy profiles for an uncatalyzed reaction and catalyzed reactions (for a positive catalyst and a negative catalyst) **N13** **N14**

Extension

* A phase is defined as a uniform portion of matter separated by other portions of matter by a boundary surface. For example, a mixture of gases is a single phase because there is no boundary surface between the different gases. However, a mixture of oil and water have a boundary surface and there are two phases.

ASL 2000 Q14
ASL 2003 Q4(a)(i)
ASL 2005 Q13
AL 2000 Paper 1 Q9

Types of catalysts

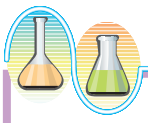
Besides considering a catalyst as positive or negative, we can also classify a catalyst as homogeneous and heterogeneous.

- Homogeneous catalyst — where the catalyst and the reactants are in the same phase*.
- Heterogeneous catalyst — where the catalyst and the reactants are in different phases.

Table 46.1 Some typical catalysts

Reaction	Catalyst	Type
$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$	vanadium(V) oxide	heterogeneous
$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$	manganese(IV) oxide	heterogeneous
$CH_3COOH(l) + CH_3OH(l) \longrightarrow CH_3COOCH_3(l) + H_2O(l)$	concentrated sulphuric acd	homogeneous
$CH_3COCH_3(aq) + I_2(aq) \longrightarrow CH_3COCH_2I(aq) + HI(aq)$	acid	homogeneous

In Topic 12 Patterns in the Chemical World, we have learnt that many transition metals and their compounds are effective catalysts. In the following activity, you are going to investigate the effect of transition metal ions on the rate of a reaction.



Activity 46.2

Catalyzing the reaction between peroxodisulphate ion and iodide ion.



Checkpoint Refer to page T39.

The activation energy for the decomposition of hydrogen peroxide in solution to oxygen and water is 36.4 kJ mol^{-1} in the presence of an enzyme and 49.0 kJ mol^{-1} in the presence of platinum.

The overall reaction is exothermic.

a) Sketch, on the same diagram, the energy profiles for reactions involving both catalysts.

b) How will the rate of the decomposition of hydrogen peroxide differ for the two catalysts at room temperature? Explain your answer.

Extension

46.8 Heterogeneous catalysis

In heterogeneous catalysis, the reactants and the catalyst are in different phases. Usually the catalyst is a solid and the reactants are either gases or liquids. Heterogeneous catalysis is by far the most important type of catalysis in chemical industry.

The Haber synthesis of ammonia

One important example of the use of solid heterogeneous catalyst is in the Haber process*. The process uses iron to catalyze the reaction between nitrogen and hydrogen:

$$N_2(g) + 3H_2(g) \overset{\text{iron}}{\rightleftharpoons} 2NH_3(g)$$

This ranks among the most important industrial reactions because ammonia is used to produce nitrogenous fertilizers*.

A heterogeneous catalyst works by providing a surface on which the reactant particles can bond. Fig. 46.18 shows how the iron catalyst functions in the Haber process.

AL 2000 Paper 1 Q9

* We have discussed the Haber process in Topic 10 Rate of Reaction, Topic 11 Chemical Equilibrium and Topic 12 Patterns in the Chemical World.

* We will discuss the production of nitrogenous fertilizers using ammonia in Unit 47.

AL 2000 Paper 1 Q9

Extension

Reference Website

Refer to the following websites for animations on heterogeneous catalytic action:

- http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/ethylene8.html (accessed May 2015)
- http://group.chem.iastate.edu/Greenbowe/sections/projectfolder/animations/Catalyst2NOO2N28.html (accessed May 2015)

The nitrogen and hydrogen molecules form bonds with iron atoms at the surface of the catalyst — we say they are *adsorbed* onto the surface. As a result, bonds in the reactant molecules are weakened and break (Stage A of Fig. 46.18).

Nitrogen and hydrogen molecules dissociate into atoms as covalent bonds break. Nitrogen and hydrogen atoms bond to the iron atoms (Stage B of Fig. 46.18).

Nitrogen and hydrogen atoms, held close together on the catalyst surface, combine to form ammonia molecules. This in turn weakens the bonds to the iron atoms at the catalyst surface, and the ammonia molecules are released — we say they are *desorbed* from the catalyst (Stage C of Fig. 46.18).

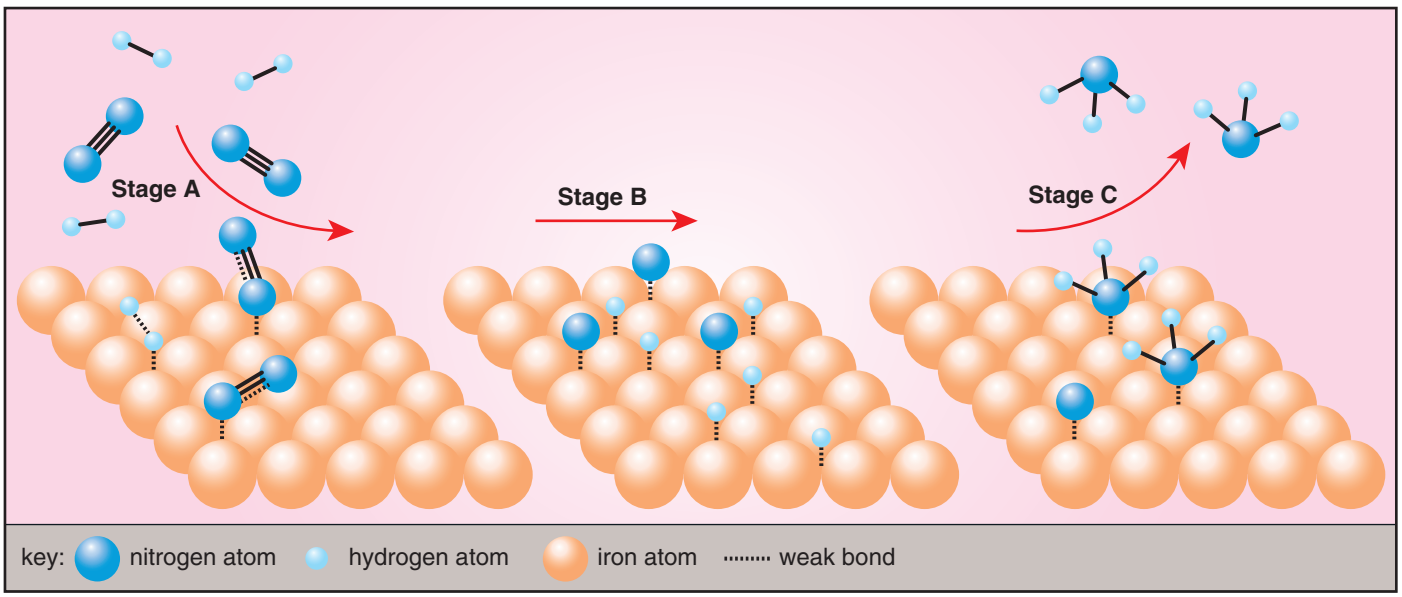


Fig. 46.18 A model for the reaction pathway for the formation of ammonia from nitrogen and hydrogen by the Haber process

It is important that the catalyst has a large surface area for contact with reactants. For this reason, solid catalysts are used in a finely divided form or as a fine wire mesh. Sometimes the catalyst is supported on a porous material to increase its surface area.

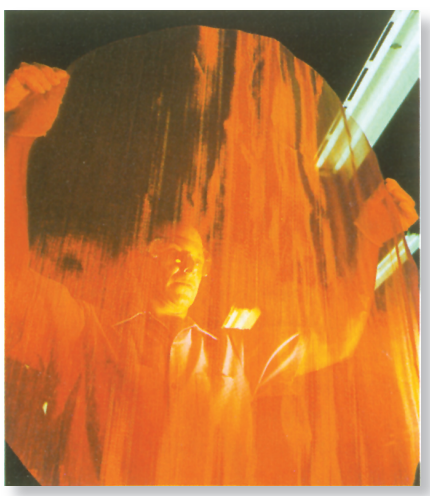


Fig. 46.19 A catalyst in the form of a gauze

adsorb 吸附　　desorb 脫附

Extension

* We have discussed the catalytic converter in Topic 10 Rate of Reaction.

Catalytic converters

The catalytic converter* is another good example of heterogeneous catalysis in action. When exhaust gas passes through the catalytic converter containing metal catalysts such as platinum (or palladium) and rhodium, harmful pollutants in the exhaust gas are converted to relatively harmless nitrogen, carbon dioxide and water.

The catalysts are supported on a honeycomb structure that allows the best possible contact between the flowing exhaust gas and the catalyst surface.

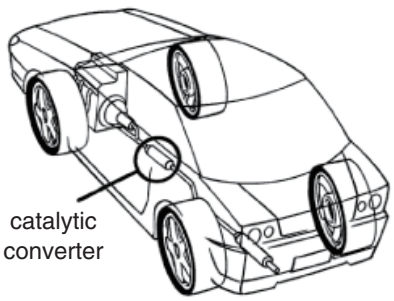


Fig. 46.20 A catalytic converter in a car



Fig. 46.21 Cutaway of a catalytic converter, showing the honeycomb catalyst structure

The catalytic converter helps to promote the following reactions:

- the reduction of nitrogen monoxide back to nitrogen:

$$2CO(g) + 2NO(g) \xrightarrow{\text{rhodium}} 2CO_2(g) + N_2(g)$$

- the oxidation of carbon monoxide to carbon dioxide:

$$2CO(g) + O_2(g) \xrightarrow{\text{platinum}} 2CO_2(g)$$

- the oxidation of unburnt hydrocarbons (e.g. octane) to carbon dioxide and water:

$$2C_8H_{18}(l) + 25O_2(g) \xrightarrow{\text{platinum}} 16CO_2(g) + 18H_2O(l)$$

octane

ASL 1999 Q10(d)

Extension



Fig. 46.22 Unleaded petrol is used in cars fitted with catalytic converters

Catalyst poisoning

Catalysts can be poisoned so that they can no longer function properly. In heterogeneous catalysis, the 'poison' molecules are adsorbed more strongly onto the catalyst surface than the reactant molecules and the catalyst becomes inactive.

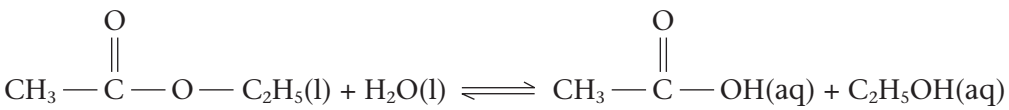
During the Haber process, it is vital that traces of sulphur and carbon monoxide are removed from the feedstocks because these will poison the iron catalyst.

Catalyst poisoning is also the reason that leaded petrol cannot be used in cars fitted with catalytic converters. This is because lead is strongly adsorbed onto the surface of the catalysts.

AL 2000 Paper 1 Q9

46.9 Homogeneous catalysis

In homogeneous catalysis, the reactants and the catalyst are dispersed in a single phase, usually a liquid. Acid and base catalyses are the most important type of homogeneous catalysis in solution. For example, the reaction of ethyl ethanoate with water to form ethanoic acid and ethanol can be catalyzed by an acid.



Another reaction that involves homogeneous catalysis is the oxidation of iodide ion by peroxodisulphate ion:

$$\underset{\text{iodide ion}}{2I^-(aq)} + \underset{\text{peroxodisulphate ion}}{S_2O_8^{2-}(aq)} \longrightarrow I_2(aq) + 2SO_4^{2-}(aq)$$

The reaction is quite slow. Both ions are negatively charged, so they are unlikely to make fruitful collisions with each other.

AL 2008 Paper 2 Q2(b)(ii)

However, if iron(II) ion is added to the reaction, the rate is much higher. The positive iron(II) ion will make effective collision with the negative peroxodisulphate ion. The iron(II) ion is oxidized to iron(III) ion which in turn oxidizes the iodide ion to iodine. In this redox reaction, the iron(III) ion is reduced to iron(II) ion. This will then continue the catalysis.

$$2Fe^{2+}(aq) + S_2O_8^{2-}(aq) \longrightarrow 2Fe^{3+}(aq) + 2SO_4^{2-}(aq)$$
$$2Fe^{3+}(aq) + 2I^-(aq) \longrightarrow 2Fe^{2+}(aq) + I_2(aq)$$

Overall reaction $2I^-(aq) + S_2O_8^{2-}(aq) \longrightarrow I_2(aq) + 2SO_4^{2-}(aq)$

N15 Ethanol from biomass

Ethanol can be produced by fermentation of plant materials. However, fermentation with yeast, the traditional method, does not convert all the sugars to ethanol. KO11 is a genetically modified bacterium developed by the microbiologist Lonnie Ingram in 1987. It converts all sugars into ethanol.

The advantage of using this bacterium to produce ethanol rather than using yeast is that a wider range of sugars can be processed. This enables biomass (such as wood waste, corn stalks and rice hulls) to be used for ethanol production.

There is an increasing tendency for farmers to grow crops such as maize to make bioethanol. However, it must be borne in mind that, in doing this, the amount of food grown for the increasing world population is not forgotten.

46.10 Enzyme catalysis

Chemical reactions that take place in living cells are catalyzed by very efficient biological catalysts called enzymes. Enzymes are usually homogeneous with the reactants and present in the same aqueous solution.

Hydrogen peroxide is a common by-product of metabolic reactions. In high concentration, it is toxic. Therefore its accumulation in cells would be harmful. Most tissues, however, contain the enzyme which catalyzes the breakdown of peroxide to water and oxygen.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

Making ethanol by fermentation

CE 2000 Paper 1 Q8(b)(iii)
CE 2001 Paper 2 Q12
CE 2002 Paper 2 Q12
CE 2004 Paper 2 Q33

The making of alcohol was one of the first chemical reactions carried out by people. The alcohol in alcoholic drinks is ethanol.

Ethanol can be produced by the fermentation* of sugars and starches. In beer making, starch is extracted from barley by soaking in hot water. The starch is then heated to 55 °C with malt. The malt contains enzymes that break down the starch to maltose ($C_{12}H_{22}O_{11}$). N15

Finally, yeast is added to the maltose. Yeast contains enzymes which break down the maltose first to glucose and then to ethanol and carbon dioxide.

DSE 2014 Paper 2 Q1(a)(ii)

$$\underset{\text{maltose}}{C_{12}H_{22}O_{11}(aq)} + H_2O(l) \xrightarrow{\text{enzymes in yeast}} \underset{\text{glucose}}{2C_6H_{12}O_6(aq)}$$

$$\underset{\text{glucose}}{C_6H_{12}O_6(aq)} \xrightarrow{\text{enzymes in yeast}} \underset{\text{ethanol}}{2C_2H_5OH(aq)} + \underset{\text{carbon dioxide}}{2CO_2(g)}$$

Wine is made by fermenting a sweet sugary liquid extracted from grapes.

* We have discussed fermentation in Topic 10 Rate of Reaction.

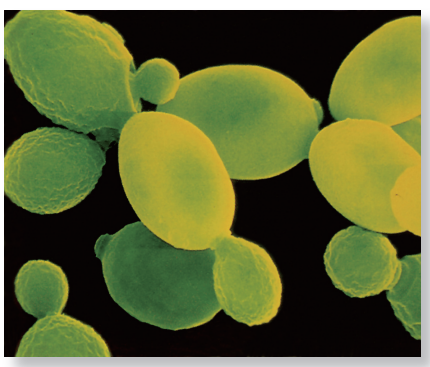


Fig. 46.23 Yeast cells

Fermentation is the slow breakdown of large organic molecules into small molecules by microorganisms in the absence of oxygen.



Fig. 46.24 Yeast floating on the liquid of a brewing fermentation vat



Fig. 46.25 The colour of red wine comes from the skin of black grapes

Fig. 46.26 shows an experimental set-up for carrying out fermentation using glucose solution. Fermentation depends on anaerobic respiration (which is a respiration without oxygen) of yeast. Yeast is a living organism and its respiration is controlled by a series of enzymes, collectively called *zymase*.

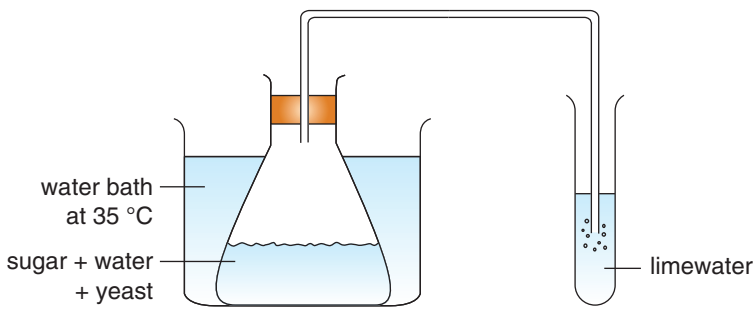


Fig. 46.26 An experimental set-up for carrying out fermentation*

* The limewater turns milky which shows that carbon dioxide is produced.

DSE 2014 Paper 2 Q1(a)(ii)

The reaction is carried out between temperatures of 25 °C and 45 °C. At higher temperatures, enzymes in yeast are denatured and stop working. It is also important that oxygen is not present because, if so, ethanol will be oxidized to produce ethanoic acid.

Fermentation can proceed until the ethanol concentration reaches about 15% by volume. At this concentration the yeast dies and fermentation stops. We can increase the concentration of ethanol by fractional distillation*.

* Fractional distillation of the fermented product of barley produces whisky which contains 37% – 40% of ethanol by volume.

CE 2003 Paper 1 Q8(a)(iii)(1)

zymase 釀酶



Fig. 46.27 Whisky and other spirits are fractionally distilled to increase their ethanol content

Table 46.2 shows sources of sugars or starches for different alcoholic drinks.

Table 46.2 Sources of sugars or starches for different alcoholic drinks

Drink	Source of sugar / starch	Drink	Source of sugar / starch
Beer	barley	Whisky	barley
Wine	grapes	Sherry	grapes

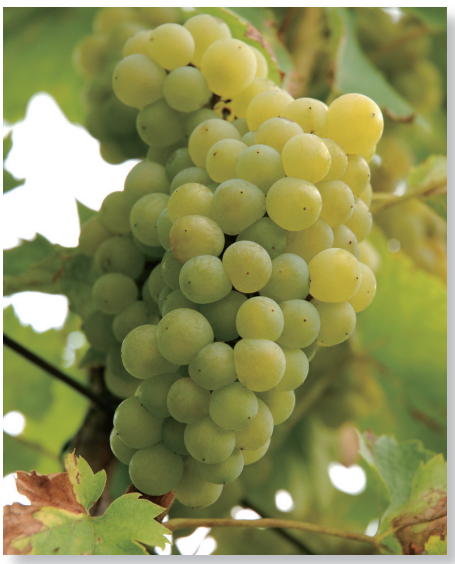


Fig. 46.28 Grapes are the raw material for making wine



Fig. 46.29 Barley is the raw material for making beer

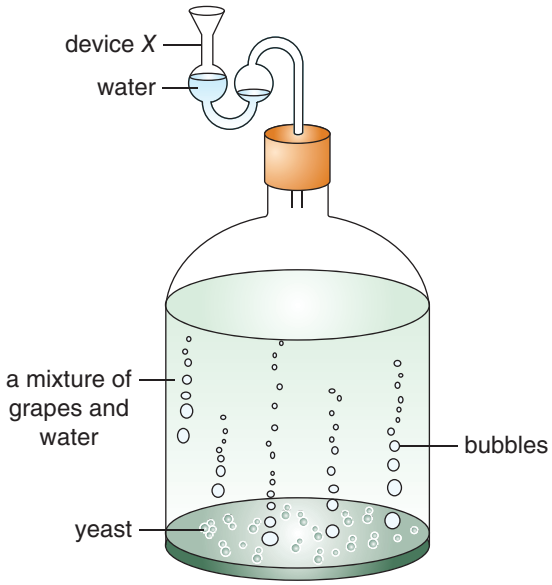
Why are enzymes so popular in chemical industry?

Enzymes are becoming more widely used as industrial catalysts because:

- they usually work at best at quite low temperatures and thus save money on heating;
- they operate in mild conditions (in water and usually close to pH 7) that do not harm fabrics or food;
- we can control them easily and stop them working by changing the temperature or pH;
- they are biodegradable and thus disposal of waste enzymes does not cause environmental problems.

Checkpoint

A mixture of grapes, water and yeast is used to produce wine in the set-up shown below.



a) Suggest ONE substance in grapes that can be converted to ethanol. Write a balanced equation for the reaction involved.

b) State the function of yeast in the production of ethanol.

c) State TWO functions of device X.

d) The ethanol concentration in the wine cannot exceed about 15% by volume. Suggest a method to increase the ethanol concentration in the wine.

CE 2003 Paper 1 Q8(a)

Checkpoint

a) Glucose

$$C_6H_{12}O_6(aq) \xrightarrow{\text{enzymes in yeast}} 2C_2H_5OH(aq) + 2CO_2(g)$$

b) Yeast contains enzymes which break down glucose to ethanol and carbon dioxide.

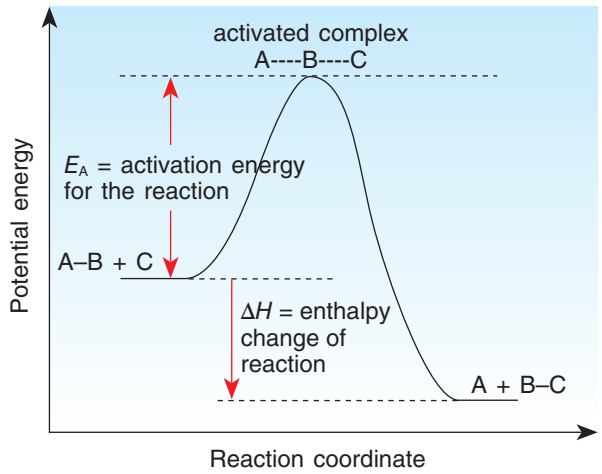
c) Prevent air from entering the jar. Otherwise the ethanol produced will be oxidized.
Prevent building up of pressure in the jar.

d) Fractional distillation

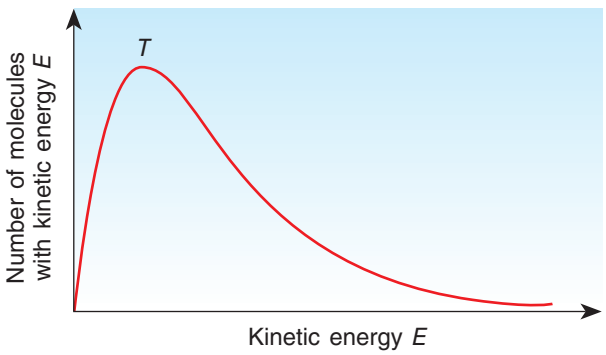
Summary

1 Activation energy is the minimum kinetic energy required by colliding particles for a reaction to occur.

2 The following diagram shows the energy profile for an exothermic reaction A–B + C ⟶ A + B–C.



3 The distribution of kinetic energy among molecules in a gas is called the Maxwell-Boltzmann distribution of energies. The curve below shows the number of molecules at each energy value at temperature T.



4 A reaction goes faster at a higher temperature because a larger proportion of the colliding molecules have the minimum energy needed to react.

5 a) The Arrhenius equation is

$$k = Ae^{-\frac{E_A}{RT}}$$

where k is the rate constant of the reaction;
A is a constant which is independent of temperature;
e is the base of the natural logarithm;
E_A is the activation energy for the reaction in J mol^{-1};
R is the ideal gas constant (i.e. 8.31 J K^{-1} mol^{-1}); and
T is the temperature in kelvin.

b) $\log k = \log A - \frac{E_A}{2.3R} \times \frac{1}{T}$

So, a graph of log k against $\frac{1}{T}$ will give a straight line of slope $-\frac{E_A}{2.3R}$. We can calculate the value of the activation energy, E_A, from the slope.

6 A positive catalyst is one that speeds up a reaction while a negative catalyst is one that slows down a reaction.

7 A catalyst works by providing an alternative pathway for the reaction to take place. A positive catalyst provides an alternative pathway with a lower activation energy. A negative catalyst provides an alternative pathway with a higher activation energy.

Extension

8 Homogeneous catalyst — where the catalyst and the reactants are in the same phase.

Heterogeneous catalyst — where the catalyst and the reactants are in different phases.

9 Enzymes are biological catalysts. They are usually homogeneous with the reactants and present in the same aqueous solution.

10 Ethanol can be produced by the fermentation of sugars and starches. Enzymes in yeast can convert glucose to ethanol and carbon dioxide.

$$\underset{\text{glucose}}{C_6H_{12}O_6(aq)} \xrightarrow{\text{enzymes in yeast}} \underset{\text{ethanol}}{2C_2H_5OH(aq)} + \underset{\text{carbon dioxide}}{2CO_2(g)}$$

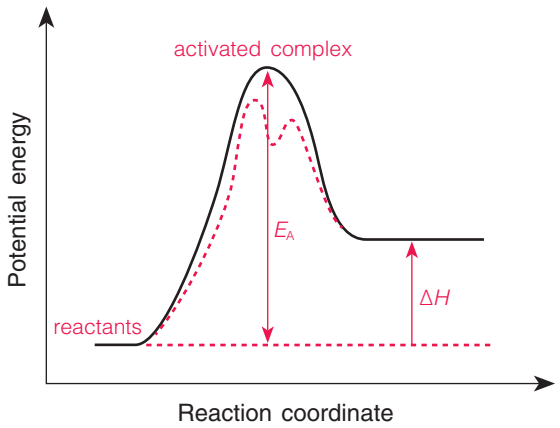
Exercise

Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 The figure below shows an unlabelled energy profile for the uncatalyzed reaction.

$CH_4(g) + H_2O(g) \longrightarrow CO(g) + 3H_2(g)$



Indicate on this profile,

a) the position of the reactants;

b) the position of the activated complex;

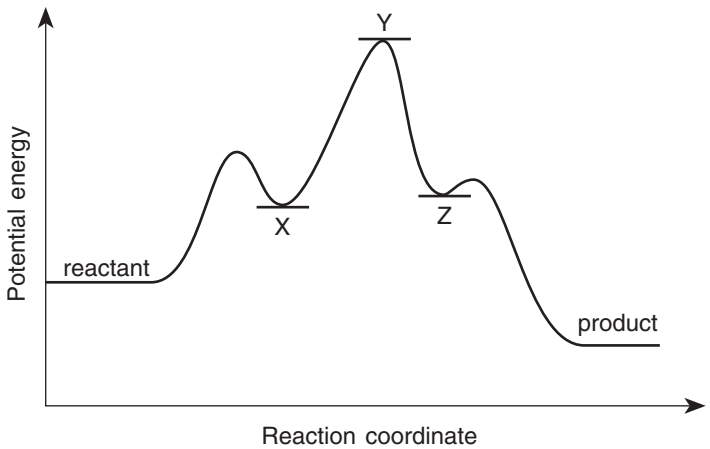
c) the enthalpy change (ΔH) for the forward reaction;

d) the activation energy for the forward reaction (E_A);

e) a possible energy profile when the reaction is catalyzed. (Use a dotted line for this.)

Extension

2 Consider the energy profile shown below for a certain reaction which takes place with three steps.



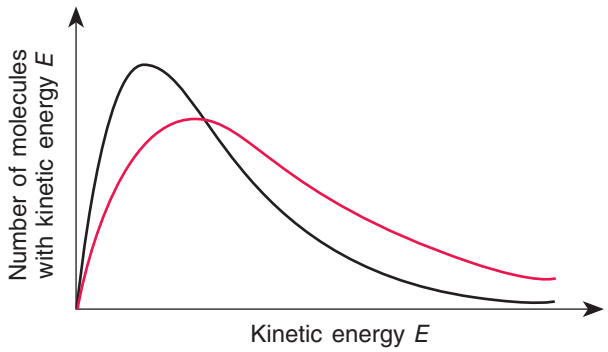
Extension

a) What are the general names for X and Y respectively?
Intermediates

b) Which is the rate-determining step? Explain your answer. X ⟶ Y
The rate-determining step is the one with the greatest activation energy.

c) Is the reaction endothermic or exothermic? Explain your answer. Exothermic
The potential energy of the product is lower than that of the reactant.

3 The diagram below shows the energy distribution of molecules at a temperature T_1.



a) Sketch on the diagram a distribution curve for the same sample of molecules at a higher temperature.

b) What does the area under either curve represent?
The total number of molecules

Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

4 The Arrhenius equation

$$\log k = \log A - \frac{E_A}{2.3R} \times \frac{1}{T}$$

relates the rate constant (k) to the activation energy (E_A) for a reaction. A is a constant and R is the ideal gas constant.

a) Comment on the relative values of the rate constant at a given arbitrary temperature T for two reactions, one of which has an activation energy of about 180 kJ mol^{-1}, the other of which has an activation energy of about 50 kJ mol^{-1}. No calculation is expected.
Reaction with a higher activation energy has a smaller k.

b) Suppose that these two different values of activation energy were for the same overall reaction at the same temperature. Suggest a way in which the conditions are different for the two reactions.
A catalyst is present in one of the reactions.

5 a) Name a catalyst for each of the following catalytic reactions.

i) Haber Process
Iron

ii) Conversion of SO_2 to SO_3 in contact process
Vanadium(V) oxide

iii) Catalytic converter in a car exhaust system
Platinum / rhodium / palladium

iv) Esterification
Concentrated sulphuric acid

v) Fermentation of glucose
Enzymes in yeast

vi) Decomposition of H_2O_2
Manganese(IV) oxide

b) Many catalysts are very expensive but their use does allow the chemical industry to operate more profitably. Give THREE reasons to explain why the use of catalysts provides economic benefits to the industry.
- Reaction proceeds faster.
- Reaction proceeds at a lower temperature.
- Energy costs are saved.

6 At 400 °C, hydrogen iodide decomposes to give hydrogen and iodine according to the following equation: Refer to page T40.

$$2HI(g) \rightleftharpoons H_2(g) + I_2(g) \qquad \Delta H = +21 \text{ kJ mol}^{-1}$$

At 400 °C, the activation energy for the forward reaction is 184 kJ mol^{-1}.

a) Sketch a labelled energy profile for the reaction.

b) Determine the activation energy for the backward reaction.
163 kJ mol^{-1}

c) State the effect of increasing the temperature on the decomposition of hydrogen iodide. Explain your answer.

d) In the presence of a gold catalyst, the activation energy for the backward reaction is 85 kJ mol^{-1}.

i) On your sketch in (a), draw the energy profile for the catalyzed reaction.

ii) Determine the activation energy for the forward reaction.
106 kJ mol^{-1}

7 Chlorofluorocarbons (CFCs) were once widely used as aerosol propellants and refrigerants. It is now known that the release of CFCs into the atmosphere will lead to the depletion of ozone in the ***stratosphere***. In the stratosphere, molecules of CFCs and O_3 absorb ultra-violet radiation and undergo decomposition to give Cl• and O atoms respectively. The following two reactions then occur:

$$Cl\bullet + O_3 \longrightarrow ClO\bullet + O_2$$
$$ClO\bullet + O \longrightarrow Cl\bullet + O_2$$

Combination of the above equations gives the overall equation for the depletion of ozone:

$$O_3 + O \longrightarrow 2O_2$$

a) Cl• can be considered as a catalyst in the depletion of ozone.

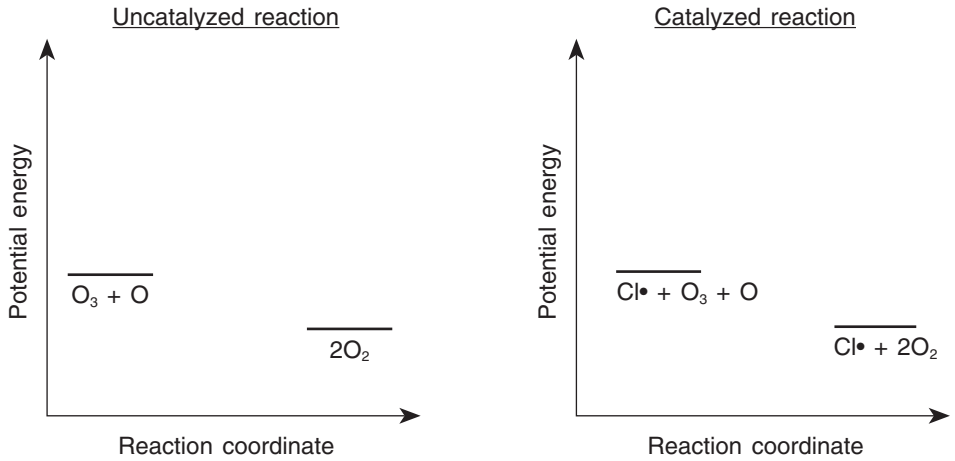
Extension

i) Is Cl• a homogenous catalyst or a heterogeneous catalyst? Explain.

ii) Sketch the energy profiles for the uncatalyzed and catalyzed reactions.

Exercise

stratosphere 同温層



b) Explain why a CFC molecule can lead to the depletion of a large number of ozone molecules in the stratosphere.

c) i) State ONE adverse consequence associated with depletion of ozone in the stratosphere.

ii) Suggest ONE substitute for CFCs, which does not cause the depletion of ozone in the stratosphere.

(HKASLE 2003)

8 Consider the following system, which comprises two single step reactions.

$$E(g) + F(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} 2G(g) \qquad \Delta H < 0$$

(k_1 and k_{-1} are the rate constants.)

a) Write the respective rate equations for the forward and backward reactions.

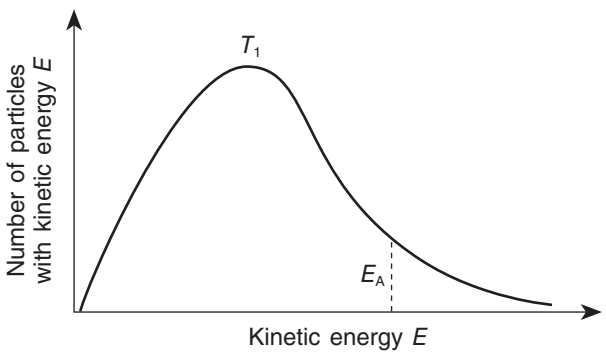
b) Sketch a labelled energy profile for the forward reaction.

c) Predict, with explanation, whether k_1 or k_{-1} will increase to a greater extent when the temperature of the system is increased.

(HKALE 2004)

9 The distribution of the kinetic energy of particles in a gas at temperature T_1 is shown below.

Refer to page T40.



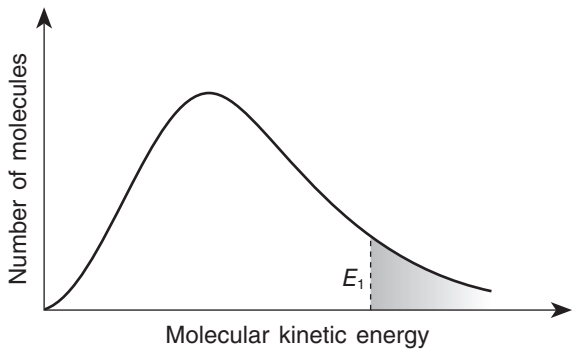
a) On the diagram above, draw the distribution of the kinetic energy of particles at a LOWER temperature, T_2.

b) Use the diagram to explain why the rate of a reaction increases with an increase in temperature.

c) Explain fully why a catalyst increases the rate of a reaction.

(Edexcel Advanced GCE (Nuffield), Unit Test 4, Jun. 2005)

10 The graph below shows the distribution of the molecular kinetic energies of a gas at a certain temperature. E_1 is the activation energy for a reaction of the gas without a catalyst.



a) Copy the graph, including the position of E_1, and mark on it the expected activation energy, E_2, of the reaction in the presence of a catalyst at the same temperature. Explain your answer.

b) Explain why the rate of a catalyzed reaction would be different from that of an uncatalyzed one at the same temperature.

(HKASLE 2006)

11 Hydrogen peroxide decomposes according to Refer to page T41.

$$2H_2O_2(aq) \longrightarrow 2H_2O(l) + O_2(g)$$

The reaction is catalyzed by solid manganese(IV) oxide, MnO_2.

a) Two experiments are carried out under the same conditions except that in one experiment the manganese(IV) oxide is a fine powder and in the other the same mass of coarse granules is used.

Which experiment would show the faster rate of decomposition? Explain your answer in term of collision theory.

b) i) Draw a labelled Maxwell-Boltzmann distribution of molecular energies in a mixture of gases at two different temperature T_1 and T_2, where T_2 is higher than T_1.

ii) Mark on your graph a suitable value for the activation energy, E_A, for the reaction between the gases. Use your drawing to account for the increase in reaction rate with increasing temperature.

iii) Use the Maxwell-Boltzmann distribution to explain why a catalyst increases the rate of a reaction at temperature T_1.

(Edexcel Advanced Subsidiary GCE, Unit Test 2, Jun. 2006)

12 2-bromo-2-methylpropane reacts with sodium hydroxide solution according to the following equation:
Refer to page T41.

$(CH_3)_3CBr(l) + NaOH(aq) \longrightarrow (CH_3)_3COH(aq) + NaBr(aq)$

A series of experiments was carried out to investigate the kinetics of this reaction.

Initial concentration of $(CH_3)_3CBr(l)$ (mol dm^{-3})	Initial concentration of NaOH(aq) (mol dm^{-3})	Initial rate of reaction (mol dm^{-3} s^{-1})
5.0×10^{-4}	3.0×10^{-2}	1.5×10^{-4}
1.5×10^{-3}	3.0×10^{-2}	4.5×10^{-4}
1.5×10^{-3}	5.0×10^{-2}	4.5×10^{-4}

a) Give the order of the reaction with respect to

i) 2-bromo-2-methylpropane;
1

ii) sodium hydroxide solution.
0

b) Write the rate equation for this reaction.

c) Calculate the rate constant of this reaction. $0.30\ s^{-1}$

Extension

d) The slowest step in the reaction is as follows:

$(CH_3)_3CBr \longrightarrow (CH_3)_3C^+ + Br^-$

Is your rate equation consistent with this information? Explain your answer.

e) In another series of experiments, the initial rate of the reaction was measured at different temperatures.

Temperature T (K)	Initial rate (mol dm^{-3} s^{-1})
300	3.35×10^{-4}
310	7.47×10^{-4}
320	1.66×10^{-3}
330	?
340	9.10×10^{-3}
350	1.66×10^{-2}

i) Determine the activation energy for the reaction by plotting a suitable graph.
$70.9\ kJ\ mol^{-1}$

ii) Estimate the initial rate of reaction at 330 K.
$3.98 \times 10^{-3}\ mol\ dm^{-3}\ s^{-1}$

(Ideal gas constant R = 8.31 J K^{-1} mol^{-1})

13 a) Explain what is meant by the following terms. Refer to page T43.

i) Rate of reaction

ii) Overall order of a reaction

b) Two gases, A and B, react according to the equation

$A + 3B \longrightarrow AB_3$

A series of experiments carried out at 298 K gave the following results.

Experiment	Initial concentration of A (mol dm^{-3})	Initial concentration of B (mol dm^{-3})	Initial rate of reaction (mol dm^{-3} min^{-1})
1	0.100	0.100	0.00200
2	0.100	0.200	0.00798
3	0.200	0.100	0.00399

i) State the order of reaction with respect to each of the reactants. Justify your answer.
1; 2

ii) Write the rate equation for the reaction between A and B.

iii) Use the experimental data from Experiment 1 to calculate the rate constant, including units.
2.00 dm^6 mol^{-2} min^{-1}

Extension

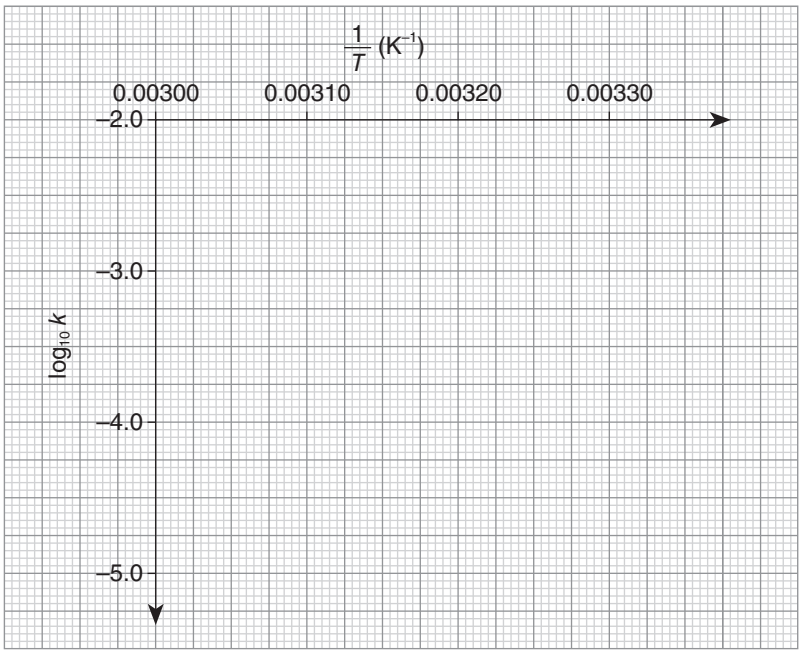
iv) Suggest a possible mechanism for the reaction between A and B, leading to the formation of AB_3. Identify the rate-determining step.

c) The rate constant, k, for the reaction in (b) was measured at different temperatures. The following data were obtained.

$\log_{10} k$	$\frac{1}{T}$ (K^{-1})
–2.70	0.00303
–3.39	0.00315
–4.09	0.00327
–4.43	0.00333

Exercise

i) Plot a graph of $\log_{10} k$ against $\frac{1}{T}$, on the grid below.



ii) The Arrhenius equation can be written

$$\log_{10} k = \text{constant} - \frac{E_A}{2.30R}\left(\frac{1}{T}\right)$$

where R = 8.31 J mol^{-1} K^{-1}.

A graph of $\log_{10} k$ against $\frac{1}{T}$ has a gradient of $-\frac{E_A}{2.30R}$.

Calculate the gradient of the graph and hence calculate the value of activation energy, E_A.

–5.67 x 10^3 K; 108 kJ mol^{-1}

(Edexcel Advanced GCE, Unit Test 5, Jun. 2006)

14 The decomposition of dinitrogen pentoxide in tetrachloromethane can be represented by the following equation:

$$2N_2O_5(\text{in } CCl_4) \longrightarrow 4NO_2(\text{in } CCl_4) + O_2(g)$$

a) Suggest an experimental method that can be used to follow the progress of the decomposition, and state the underlying principle of the method.

b) The rate constant for the decomposition at 332 K was found to be 5 times that at 318 K. Calculate the activation energy for the decomposition.

(HKALE 2004)

Extension

15 Consider the acid-catalyzed reaction:

$$X(aq) + Y(aq) \xrightarrow{H^+(aq)} Z(aq) \qquad \Delta H < 0$$

The following mechanism was proposed:

$$X(aq) + H^+(aq) \rightleftharpoons XH^+(aq) \qquad \text{(fast)}$$

$$XH^+(aq) + Y(aq) \longrightarrow Z(aq) + H^+(aq) \qquad \text{(slow)}$$

Based on this mechanism,

a) sketch a labelled energy profile, and

b) deduce the rate equation.

(HKASLE 2007)

16 Many transition metals and their compounds are useful catalysts. By choosing a suitable example in each case, illustrate and explain

a) what a homogeneous catalyst is;

b) what a heterogeneous catalyst is.

16 a) Iron(II) ion is used as a catalyst in the oxidation of iodide ion by peroxodisulphate ion.
Iron(II) ion is a homogeneous catalyst as the catalyst and the reactants are in the same phase.
b) Vanadium(V) oxide is used as a catalyst in the reaction between sulphur dioxide and oxygen gases in the Contact process.
Vanadium(V) oxide is a heterogeneous catalyst as the catalyst and the reactants are in different phases.

17 Nitrogen monoxide reacts with hydrogen at 500 °C as in the equation below. Refer to page T45.

$$2NO(g) + 2H_2(g) \longrightarrow N_2(g) + 2H_2O(g)$$

A series of experiments was carried out to investigate the kinetics of this reaction. The results are shown in the table below.

Experiment	[NO] (mol dm^{-3})	[H_2] (mol dm^{-3})	Initial rate (mol dm^{-3} s^{-1})
1	0.10	0.20	2.6
2	0.10	0.50	6.5
3	0.30	0.50	58.5

a) i) For each reactant, deduce the order of reaction. Show your reasoning.
2; 1

ii) Deduce the rate equation for this reaction.

iii) Calculate the rate constant, k, for this reaction. State the units for k.
1 300 dm^6 mol^{-2} s^{-1}

Exercise

b) Nitrogen monoxide, NO, is involved in formation of ozone at low levels and the breakdown of ozone at high levels.

i) In the lower atmosphere, NO is produced by combustion in car engines. Ozone is then formed following the series of reactions shown below.

$$NO(g) + \frac{1}{2}O_2(g) \longrightarrow NO_2(g)$$

$$NO_2(g) \longrightarrow NO(g) + O(g)$$

$$O_2(g) + O(g) \longrightarrow O_3(g)$$

- Write the overall equation for this reaction sequence.
- Identify the catalyst and justify your answer.

Extension

ii) In the upper atmosphere, NO removes O_3 by the following reaction mechanism.

$NO(g) + O_3(g) \longrightarrow NO_2(g) + O_2(g)$ slow

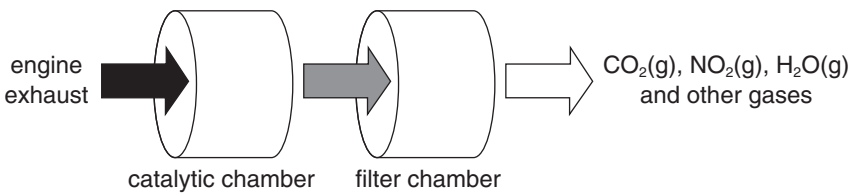
$O(g) + NO_2(g) \longrightarrow NO(g) + O_2(g)$ fast

Suggest the rate equation for this process. Explain your reasoning.

(OCR Advanced GCE, Unifying Concepts, Jan. 2007)

18 The exhaust of heavy-duty diesel engines contains a significant amount of particulate matter (PM) and harmful gases such as nitrogen oxides. A Continuously Regenerating Trap (CRT®) is a device which is designed for use in exhaust systems of buses and lorries running on diesel with low sulphur content to remove PM and some of the harmful gases.

The diagram below shows how a CRT works:



a) i) With the help of chemical equations, explain why nitrogen oxides are present in the engine exhaust.

ii) State ONE harmful effect of nitrogen oxides on the environment.

b) Carbon monoxide and hydrocarbons are two other harmful gases present in the engine exhaust.

Use chemical equations to show how these two gases can be removed in the catalytic chamber of a CRT.

c) A CRT is an automated, self-regenerating device which does not require cleaning of the filter. In a CRT, PM is trapped onto the filter and is then oxidized by one of the harmful gases to less harmful products.

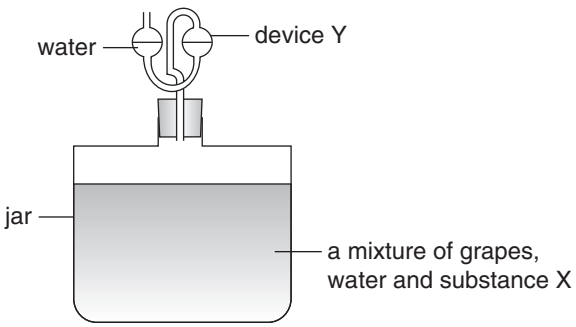
i) Which element is the most abundant in PM?

ii) With the help of chemical equation(s), describe how PM trapped on the filter of a CRT can be removed. Hence explain why the filter need NOT be cleaned.

d) Suggest why buses and lorries equipped with CRT should NOT run on diesel with high sulphur content.

(HKALE 2004)

19 A mixture of grapes, water and substance X is used to produce wine in the set-up shown below:



a) The wine contains ethanol.

i) State ONE substance in grapes that can be converted to ethanol. Write the chemical equation for the reaction involved.

ii) Suggest what X may be. State its function in the production of ethanol.

b) State TWO functions of device Y.

c) i) Explain why the concentration of ethanol in the wine cannot exceed a certain level (about 18% by volume).

ii) Suggest a means to increase the concentration of ethanol in the wine to a level higher than 18% by volume.

d) Explain why a glass of wine turns sour upon standing in air.

(HKCEE 2003)

Extension

20 Write an essay on catalysis.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

(HKASLE 2005)